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UNIVERSITY OF TORONTO

DEPARTMENT OF ENGINEERING SCIENCE

The Wobble Dilemma: Rethinking Convertible Laptop Hinges – Design Report

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1. Introduction - Add-On Hinge Mechanism Design for a 2-in-1 Laptop

This project develops a compact hinge add-on (HAO) mechanism to improve screen stability in convertible laptops during stylus use. The primary users are first year EngSci students who rely on a stable digital writing surface during lectures. Current 2-in-1 hinges cannot maintain common writing angles under optimal pen forces leading to inefficient use of the laptop. This report presents the requirements, design concepts, CAD development, prototypes and testing used to create a portable, lightweight solution.

2. Overview Interpretation Of The Laptop Design Hinges (Design brief):

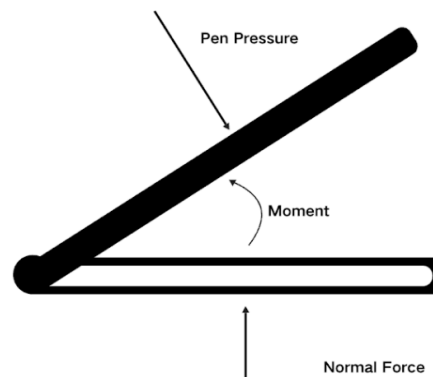
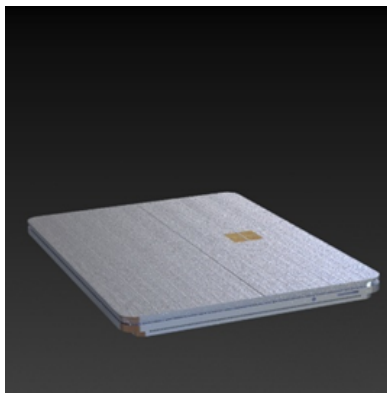
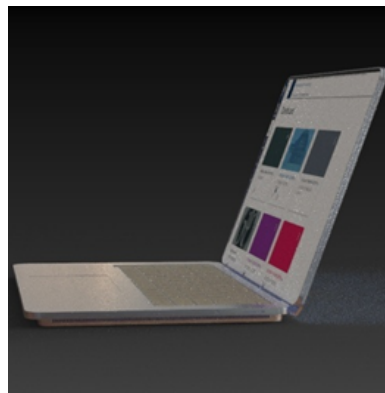


Figure 1- Free-body diagram showing how pen pressure produces a moment about the hinges, opposed by the normal force from the laptop.



a. Laptop closed



b. Typing configuration



c. Writing configuration

Figure 2 - Three operating modes of a 2-in-1 laptop. The writing configuration is the focus of this project due to the instability caused by pen pressure forces.

Current 2-in-1 hinge support systems cannot withstand typical writing forces, resulting in screen wobbles, reduced precision and compromised user posture (*Figure 1*). For first year engineering science students who rely on digital notetaking and frequent transition between laptop and tablet modes (*Figure 2*), this instability creates wasted efficiency and usability.

Evidence from classroom (McLennan and SF1105) measurement, which can be found in the *appendix F*, confirms that users required adjustable but rigid support within writing angles of 20°-50°. Existing hinges do not provide adequate torque or resistance to maintain such angles under downward or lateral pen pressure. These findings confirm the primary stakeholders priorities:

- ❖ Ergonomics: Reduce physical strain and support long study hours
- ❖ Functionality: Reliably performance in laptop and tablet mode, with improved stability during stylus use.
- ❖ Portability and aesthetics: Remain as lightweight, practical for daily use, and visually appealing.

Original NGOs of the Design brief:

Need: Students need a HAO that eliminates screen instability in convertible 2-in-1 laptops during stylus based writing, ensuring a reliable, comfortable tablet-mode experience.

Goal 1– Range of motions and hinge performance:

A stable hinge requires both rigidity and a smooth motion. To prevent vibrations and a comfortable writing experience. During stylus use, even small angular deflections produce notable vibrations that disrupt writing. The HAO must provide high rigidity while allowing a smooth rotation through 0-360°.

1.1 Resisting writing force

- ❖ Stylus writing applied forces of up to $433 \pm 92\text{g}$ (Appendix C). The hinge must not deform under the max expected writing force.

1.2 Limit screen oscillations

- ❖ The hinge must limit vibration to within $\pm 0.2^\circ$ during stylus use.

1.3 Durability benchmark

- ❖ The hinges should meet industry standard BHMA A1561.1, 350000+ cycles for certification.

Goal 2- Ergonomics

Neutral posture reduces fatigue during prolonged note-taking. Tablet mode often forces students to lean forward or overextend their wrists. The HAO must support ergonomics writing angles that minimize strain.

2.1 Adjustable writing angles

- ❖ The hinge must be able to have stable positions within the 20°-50° range.

2.2 Reduce postural deviation

- ❖ The hinges must allow the tablet to be in a position that minimizes the forward lean. (Joints angles near $90^\circ \pm 10^\circ$)

Goal 3- Portability

Accessories used daily must be lightweight and compact. Bulkier products decrease portability and discourage use.

3.1 Fit within bounding box

- ❖ The usable space for the prototype is 35x25x14.5 cm.

3.2 Weight limit

- ❖ The add-on must weigh no more than 300g.

Reframing of design brief:

During the diverging stage, we found that the splartz was too broad, and we needed to scope in. The two key things we added to constrain our design are as follows:

Objective	Description
The solution must be compatible with the Lenovo Yoga i7 laptop.	In our original design brief, the model of laptop the solution must fit on was never specified, and it was implied that it should fit on any laptop. In practice, this is not very feasible in the early stages of prototyping. To scope in, we decided to focus only on solutions that fit the Lenovo Yoga i7 since it is the laptop we have access to for testing our prototypes.

The material selected for the solution must support the writing pressure supplied while maintaining manufacturability.	When we initially wrote the design brief, we decided to neglect material selection in our NGOs, leaving it up to the designer. Now, as the designers, it was our job to worry about material as it directly impacts the weight and torque it supplies.
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Note that though we have not removed ergonomics as a goal, it will not be a requirement we evaluate since all solutions were designed to meet this requirement, and it would not be worthwhile to compare identical metrics.

3. Performance Requirements for a Stylus-Stable Hinge Add-On

We used a pairwise comparison analysis (*Appendix A*) to systematically compare each requirement to determine its relative importance for evaluation.

1. Material selection for durability and resistivity to pen pressure

Criterion: Screen stability is governed by stiffness and local deformation at the pivot. Material with high yield strengths maintain their stiffness over life meeting the series vibration limits. However, such materials may be too difficult to form, and better alternatives can be found that reach the maximum deformation requirement and have good manufacturability. The chosen material should take into consideration the manufacturability of the part, this should be considered after the previous requirements have been met.

Metric: Static stiffness requirements of deflection $\leq 0.775\text{mm}$ under 4.25 N , force is applied to the point on the screen furthest from the pivot(s) being supported. (*Appendix C*)

Evaluation : CAD FEA analysis can be used to determine maximum deflection of each material tested.

2. Spatial Volume constraints for backpack compatibility

Criterion: Primary and secondary research was conducted (*Appendix D*) to find realistic spatial constraints for portable laptop accessories. The HAO must fit within the minimum internal volume of a typical student backpack so the laptop can be transported normally with the add on (*Figure 3*).

Metric: Maximum external dimensions: $\leq 35\text{cm (L)} \times 25\text{cm (W)} \times 14.5\text{cm (H)} = 12687\text{cm}^3$

Evaluation: By creating a full scale CAD model of the HAO, we can place the mockup into the measured pink boundary box. Verify that the assembly fits without forcing or deforming the backpack. The design is considered portable if it fits within the constraints continuously in one piece.

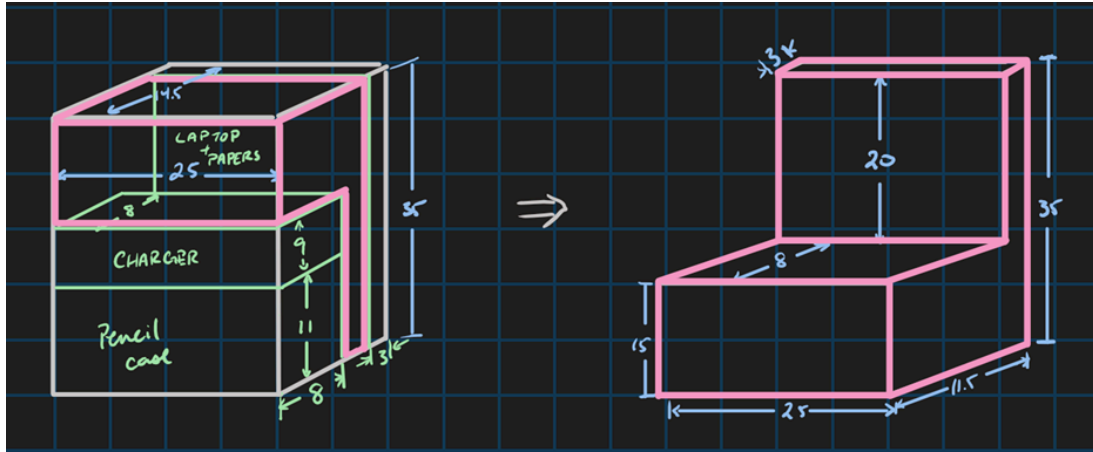


Figure 3 - Calculated bounding-box for possible solutions. Note that this is not a suggestion for the shape, but rather a shape to allow choice between a flat and/or boxy form factor.

3. Time and Installation steps — [Video](#)

Criterion: The HAO must be attachable to the laptop using an intuitive process that requires minimal time, no specialized tools and no risk of damaging the laptop. Primary user research (*Appendix G*) showed that students prefer accessories that are attached quickly between classes or work sessions.

Metric: Installation time (seconds) measured from start to completion ≤ 60 seconds. Number of steps required to attach the add-on ≤ 5 .

Evaluation: The usability test will be conducted in a video, where a team member will showcase the time to complete installation and the number of steps taken to attach the designs onto a 2-in-1 laptop. The design passes if the metrics are within range.

4. Weight

Criterion: The HAO is meant for students who carry their laptops around every day, so it needs to stay light. If the weight increases too much, the laptop becomes harder to bring to classes.

Metric: Increase in weight after adding the hinge ≤ 348 g (noted in the Design Brief). Weight measured using a digital scale.

Evaluation: We will weigh the laptop before and after the hinge is added. If the extra weight is 348 grams or less, the design meets the requirement.

4. Creating Divergent Hinge Concepts:

I-Main Idea- The Bice Chair-Beach chair

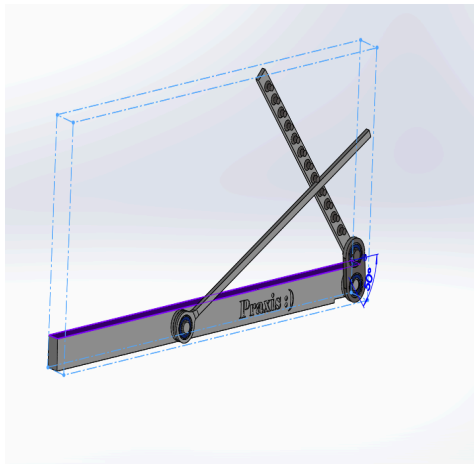
This mechanism seen in *figure 4* consists of two laptop-mounted side plates, two bearings that lock the screen at fixed angles and a pivot arm that rotates 5 inches horizontally from the rotation axis. The top side plate rotates around a bearing and contains slots to wedge the pivot arm between creating fixed angles. This allows the user to have customization to their preference and maximize the number of available positions while meaning sufficient clearance for reliable engagement.



a. Reference design



b. Initial prototype



c. CAD iteration



d. Final render

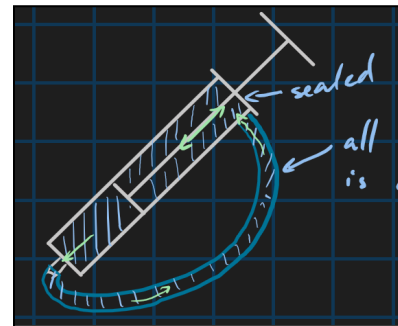
Figure 4 - Demonstrates the progression from idea to final prototype of the Bice chair

II Syringe-Like Laptop Hinges

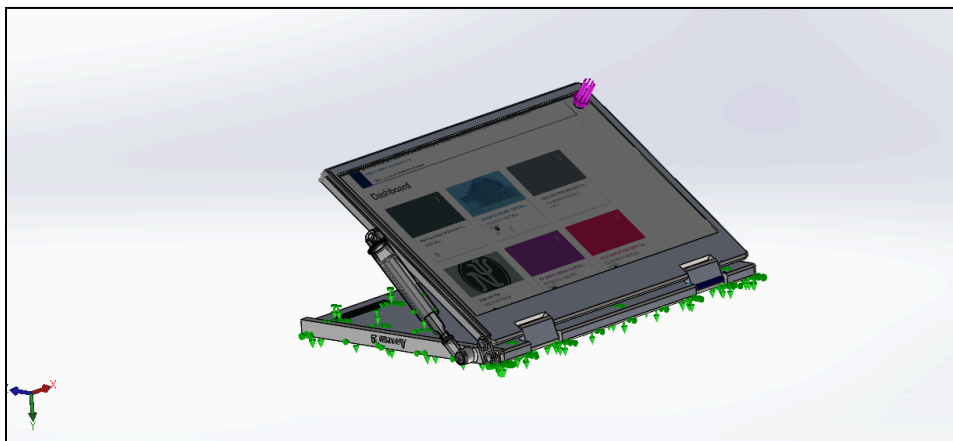
This hinge operates as a syringe-style piston containing air or hydraulic fluid that moves through small orifices during screen rotation. When the flowpath is opened, the piston moves freely allowing smooth rotation, when restricted fluid resistance increases, creating a “locked” hinge. This mechanism mirrors real-world fluid damping systems such as refrigerator door dampers or pneumatic struts, but is miniaturized and adapted to provide steady support and controlled motion for a laptop display observed in *figure 5*. It provides smooth rotation and a stable screen support during stylus use. Since the damping force depends on the fluid’s thickness and the size of the opening, we can easily fine-tune how stiff the rotation feels.



a. Reference design



b. Drawing

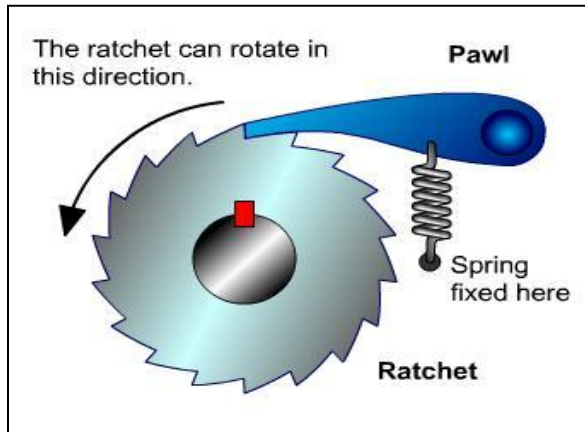


c. CAD iteration mounted to laptop

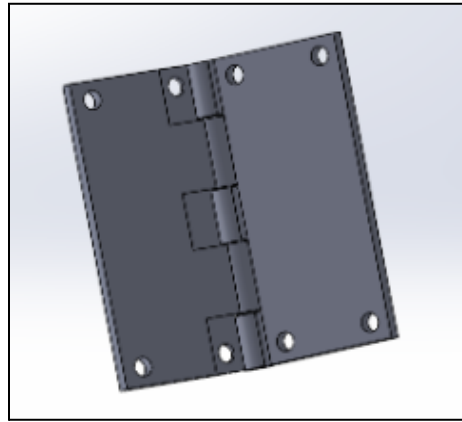
Figure 5 - Depicts the progression from idea to final prototype of the syringe design

III. Ratchet Mechanism

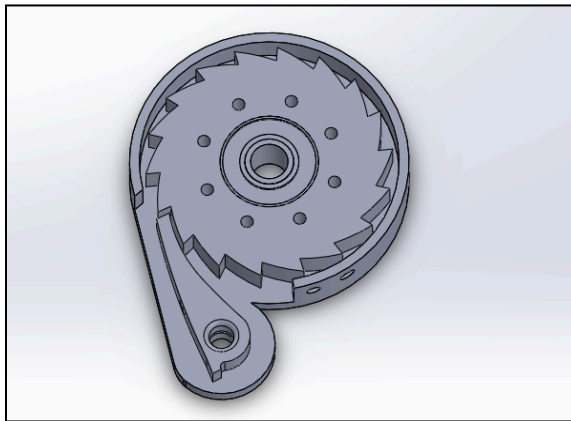
This concept as observed in *figure 6* uses a miniature ratchet and pawl mechanism built into the hinge. As the screen rotates, the ratchet wheel advances tooth-by-tooth and the pawl locks into each tooth to hold the screen at fixed angles. As the number of teeth increases, so does the variation of fixed angles.



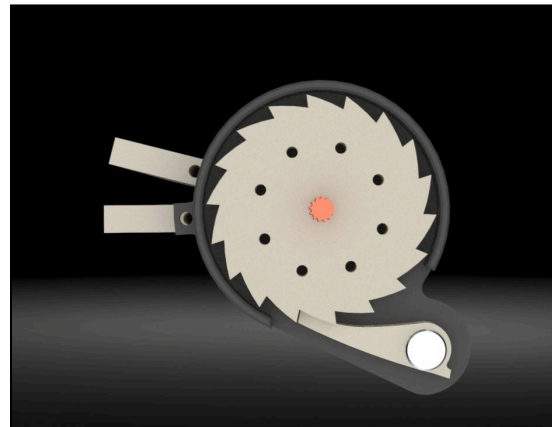
a. Reference design



b. Iteration outer design



c. Iteration ratchet mechanism



d. Final render

Figure 6 - Shows the progression from idea to final prototype of the ratchet system

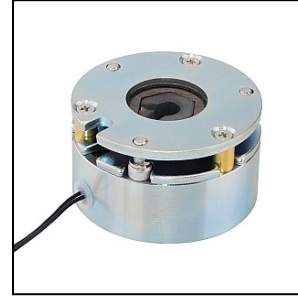
IV. Electromagnetic Hinge System

This concept uses an integrated miniature electromagnetic brake as seen in *figure 7* to generate an adjustable holding torque. When current flows through the coil, the resulting magnetic field produces a resistive torque, similar to industrial electromagnetic clutches. The torque strength depends on the magnetic and coil current. Our model describes how the electromagnet generates resistive torque under different motions conditions.

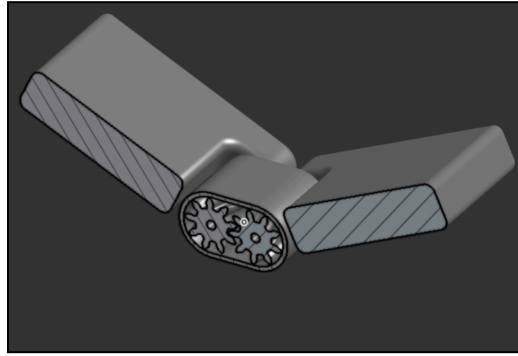
According to the formula this braking torque directly corresponds to holding torque that prevents the laptop screen from moving when stylus force is applied.

$$T = \frac{k_1 \omega}{\left(1 + \frac{k_2 \omega^2 + k_3 \omega^4}{1 + k_4 \omega + k_5 \omega^3}\right)^2}$$

a. Mathematical model



b. Electromagnetic brake

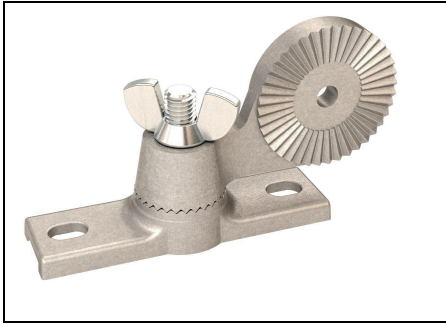


c. CAD Iteration

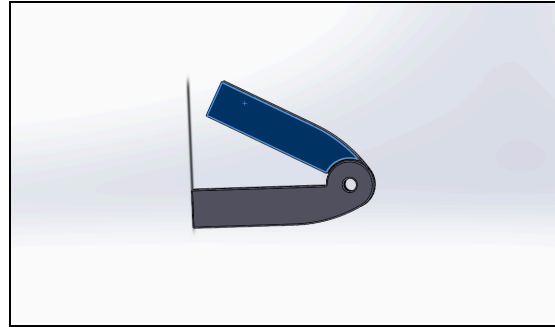
Figure 7 - showing the progression from idea to final model of the electromagnetic hinge idea

V. Screw

This concept as seen in *figure 8* uses a tighten screw to control rotation at the hinge. Two concentric gear-like plates form the pivot point. Loosening the screw allows the plates to rotate freely, while tightening it clamps them together allowing for a fixed angle. The number of teeth on the plates determines the available discrete positions, creating a simple, and reliable mechanism that is controlled by screw tension.



a. Ratchet reference design



b. Final CAD Model

Figure 8 - Showing the progression from idea to final prototype of the syringe design

Objective Met	Bice chair	Syringe	Ratchet	Electromagnet	Screw
Rigid screen support	✓	✓	✓	✓	✓
Full motion	✓	✓	✓	✓	✓
Stops at ergonomic angles (20–50°)	✓	✓	⚠ Need More Teeth (Redesign)	✓	⚠ Need More Teeth (Redesign)
Compact & lightweight	✓	⚠ Need improvement	✓	✓	✓
Durability	⚠ Need improvement	✓	⚠ Need improvement	⚠ Need improvement	⚠ Need improvement

5. Engineering Analysis of the Syringe Mechanism

Through iteration and prototyping we found that the syringe was the clear solution when compared to the other four solutions. Its unique design, reliability with dual design and one time install makes it stand out as an add on, simplifying the user experience.

To further analyze the syringe design, we considered the loads it would be under using an FEA using Solidworks simulations. For these simulations, vertical displacement at 35 degrees, the average between the maximum and minimum ideal angles, were compared to the visual displacement of the human eye of ± 0.2 degrees (*Appendix C*). Using these angles, we found the maximum theoretical vertical displacement of the screen to be 0.775 mm and would compare this to the simulations (Seen in *appendix D*)

I. Material Considerations and FEA

Material selection is critical in FEA analysis, since a different young's moduli affects buckling resistance according to:

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

Where L is length, E is the young modulus, π is a constant and I is the area moment of inertia. We initially started with a plastic with low E, Acrylonitrile Butadiene Styren (ABS) due to its ease of manufacturing, adequate strength and affordability and since it responds well to thermoforming [*Source 10*]. Parts exceeding the requirement of 0.775mm displacement, was reconsidered.

Since our design would be subjected to repeated strong forces, our research into material design was scoped for how often it is used and primarily the material strength which led us to ABS. We only modified materials for parts that required higher strength, as plastics are more easily manufacturable for parts not subjected to repeated stress. Critical moving parts, like the gear and the shaft on the ratchet are made of metals. For all our studies, a point source force representing the contact between the screen and a pen of 4.25 N was considered directly on the syringe to see the stress and displacement on both the pivot arm and the slotted support. Aspects such as bearing stiffness and material coefficient of friction were also considered in the study as seen in *appendix C*. For the FEA with ABS, this can be found below as *figure 9*:

Iteration 1: All Parts Are Composed Of The Material ABS

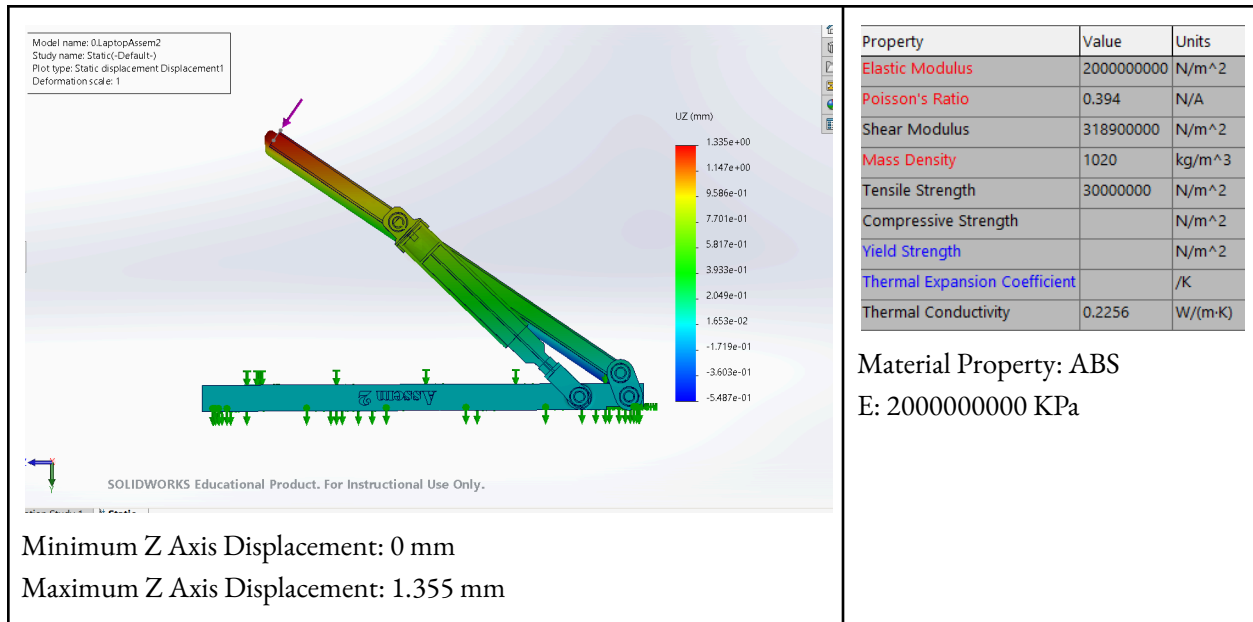


Figure 9 - Depicts the load simulation of 4.25N on the Bice chair and the resulting displacement using section clipping on a scale of no displacement (Blue) to maximum displacement (Red).

As a result of this study, we discovered that a few parts must be reconsidered for this design: the two parallel plates to the laptop subjected to the most load and the connection joints to the syringe required stronger connections. Currently, these parts did not pass the vertical displacement and needed to be optimized. This led us to ABS PC, a mix of ABS and polycarbonate, described as the toughest plastic [Source 7] as it maintains dimensional stability (See *appendix B* for more information). This iteration can be seen below with *figure 10*:

Iteration 2: Failing Materials Change

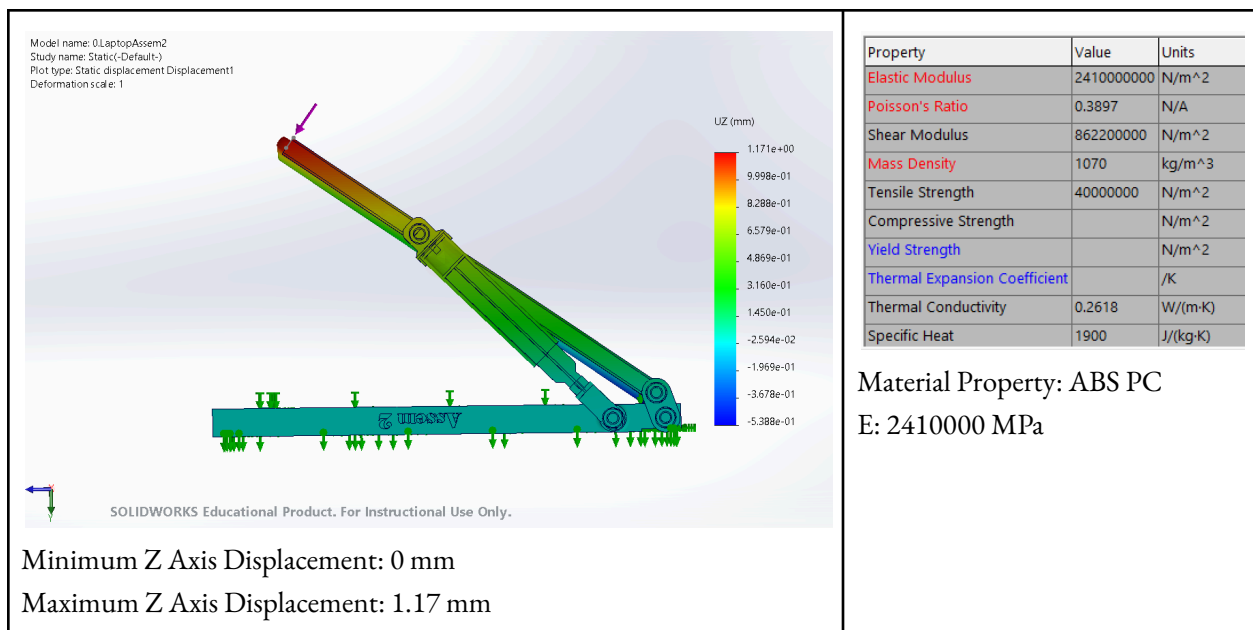


Figure 10 - Depicts the load simulation of 4.25N on the syringe given the new changes of material to connection joints. This part has a max displacement of 6.37mm which still exceeds the required displacement of 0.775mm.

This difference of max displacement of only 0.238mm led us to decide that the best material should not be a plastic due to the minimal change of displacement, instead a stronger metal should be used. Whilst this makes manufacturability more difficult, the main splartz relies on resistance and thus the decision for a metal such as stainless steel ferritic was decided. This specific type was used due to its high young's modulus value and its common use, being the second most used stainless steel after austenite [Source 11]. Its superior stiffness, excellent fatigue resistance, and ability to maintain dimensional stability under repeated loading make it ideal for a hinge mechanism that experiences frequent high-pressure stylus contact [Source 8]. The final iteration can be seen below as figure 11

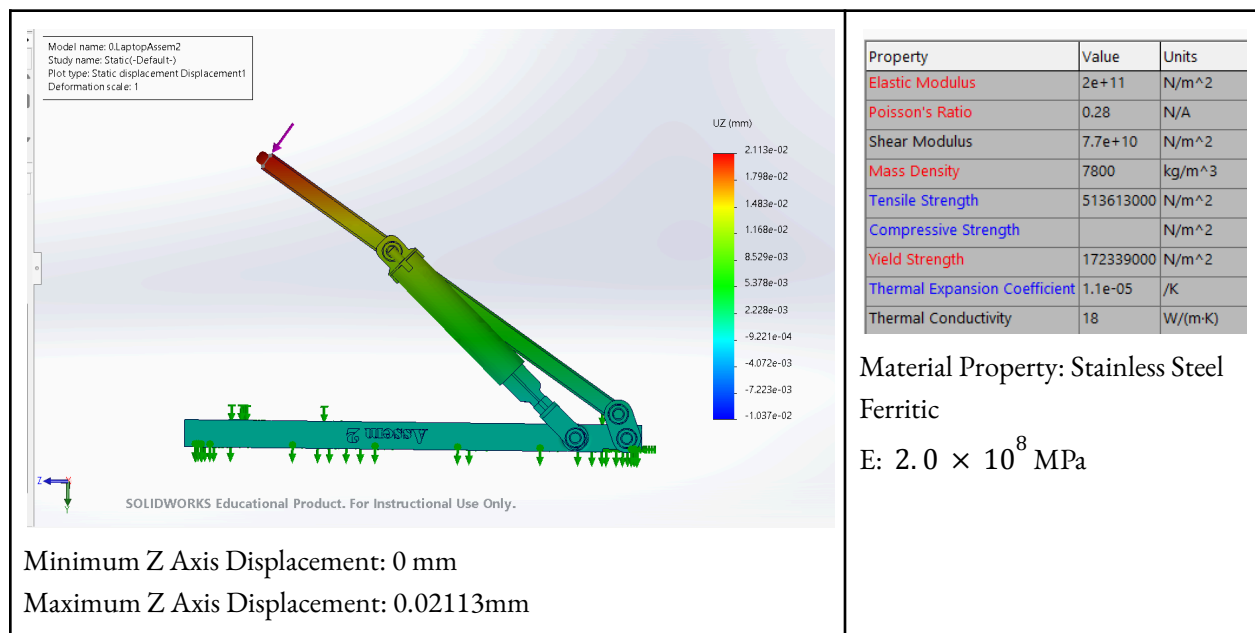


Figure 11 - Depicts the final load simulation of 4.25N on the syringe with the new changes to the materials. Unlike the iterations with plastic, this design meets the required displacement of 0.775mm and is our final design for the Bice chair.

Despite the fact we have two syringes for each side, we decided to test if the design would work at maximum force - the top right corner as seen in the figure below. This is the maximum force region because the laptop display behaves like a cantilevered plate in which the farther a point is from the pivot/support, the larger the bending moment acting on it:

$$M = F_a \times d$$

Even at a point force at its maximum location, the maximum displacement is 0.0262 mm which is way below our requirement. The regions that previously exceeded the given displacement, now are below it as seen in figure 12 below:

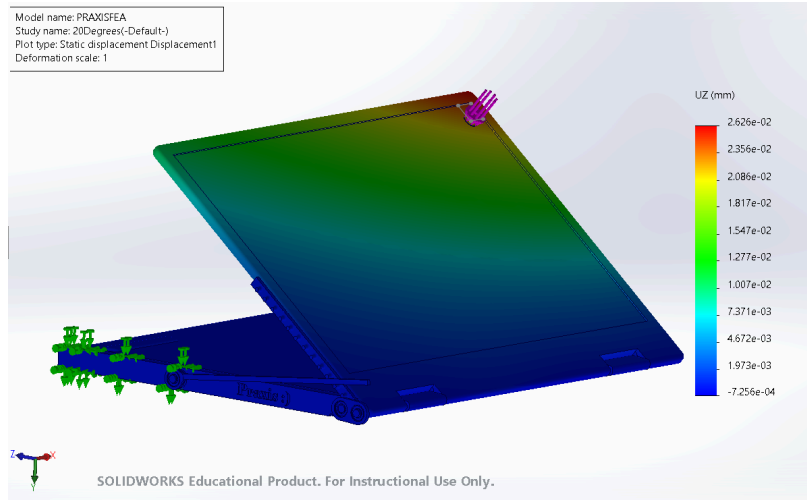


Figure 12 - Demonstrates the resultant displacement when the point force is applied at its maximum location (Represented by pink arrows at the top right screen).

II. Measurement Matrix

Please see *appendix D* for Material Deformation metric and E for volume for more information about determining each requirements metric. Note that the use of “*” Means the solution type has two parts to it. For example, the syringe solution has a syringe on both sides.

Requirement	Bice Chair	Syringe*	Ratchet	Electromagnet*	Screw*
Material selection for durability and resistivity to pen pressure (mm)	0.0774	0.0211	0.01485	0.0582	0.032
Volume Closed (cm ³)	75.9	176.0	218.7	60.5	33.6

Time and Steps (seconds - steps)	51.08s 4 steps	47.08s 5 steps	17.33 s 1 step	17.95s 2 steps	NA
Weight (g)	16.63	120.02	97.05	81.64	36.28

III. Pugh Charts

Ratchet (Baseline)

Requirement ↓ \ Prototype →	Ratchet (Baseline)	Bice Chair	Syringe	Electromagnet	Screw
Material (durability / pen resistance.)	0	- (0.0774 > 0.0149)	- (0.0211 > 0.0149)	- (0.0582 > 0.0149)	- (0.032 > 0.0149)
Volume closed (cm ³)	0	+ (75.9 < 218.7)	- (176 > 218.7)	+ (60.5 < 218.7)	+ (33.6 < 218.7)
Time & Steps	0	- (51.08 s / 4 steps > 17.33 s / 1)	- (47.08 / 5 steps > 17.33 / 1 step → -)	+ (17.95/2 < 17.33/1)	NA
Weight (g)	0	+ (16.63 < 97.05)	- (120.0 > 97.05)	+ (81.64 < 97.05)	+ (36.28 < 97.05)

INTERPRETATION:

- ❖ Best design for durability
- ❖ Quite large compared to other options
- ❖ Good design for assembly time
- ❖ Quite heavy compared to other options

Electromagnet (Baseline)

Requirement ↓ Prototype →	Electromagnet (Baseline)	Bice Chair	Syringe	Ratchet	Screw
Material (durability / pen resistance)	0	- (0.0774 > 0.0582)	+ (0.0211 < 0.0582)	+ (0.0149 < 0.0582)	+ (0.03 < 0.0582)
Volume closed (cm ³)	0	- (75.9 > 60.5)	- (176 > 60.5)	- (218.7 > 60.5)	+ (33.6 < 60.5)
Time & Steps	0	- (51.1/4 > 17.9/2)	- (47.1/5 > 17.9/2)	0	NA
Weight (g)	0	- (16.63 < 81.64)	- (120.02 < 81.64)	- (29.31 < 81.64)	+ (36.3 < 81.64)

INTERPRETATION:

- ❖ *Model for electromagnet is very high level, metrics obtained may not be completely accurate
- ❖ Reasonable design for durability
- ❖ Good size compared to other options
- ❖ Good design for assembly time

- ❖ Light compared to other options

Screw lock (Baseline)

Requirement ↓ \ Prototype →	Screw Lock (Baseline)	Bice Chair	Syringe	Ratchet	Electromagnet
Material	0	- (0.0774 > 0.032)	+ (0.021 < 0.032)	+ (0.0149 < 0.032)	- (0.06 > 0.032)
Volume closed	0	- (75.9 > 33.6)	- (176 > 33.6)	- (134.7 > 33.6)	- (61 > 33.6)
Time & Steps	0	NA	NA	NA	NA
Weight	0	- (16.63 < 36.28)	- (120.02 > 36.28)	- (29.31 < 36.28)	- (81.64 < 36.28)

INTERPRETATION:

- ❖ Reasonable design for durability
- ❖ Most compact design
- ❖ Assembly time was not tested for this design
- ❖ Light compared to other options

Bice Chair (Baseline)

Requirement ↓ \ Prototype →	Bice Chair (Baseline)	Syringe	Ratchet	Electromagnet	Screw
Material (durability / pen resistance)	0	+	+	+	+
		(0.0211 < 0.0774)	(0.01485 < 0.0774)	(0.0582 < 0.0774)	(0.032 < 0.0774)
Volume Closed (cm ³)	0	-	-	+	+
		(176 > 75.9)	(218.7 > 75.9)	(60.5 < 75.9)	(33.6 < 75.9)
Time & Steps (s / steps)	0	-	+	+	NA
		(47.08 / 5 > 40 s / 4 steps)	(17.33 / 1 < threshold)	(17.95 / 2 < threshold)	
Weight (g)	0	-	-	-	-
		(120.02 > 16.63)	(29.31 > 16.63)	(81.64 > 16.63)	(36.28 > 16.63)

INTERPRETATION:

- ❖ Syringe only scores positive for material durability, while it fails all other requirements, volume, time and weight.
- ❖ Ratchets and electromagnet excel in volume and time but are too heavy. All prototypes fail for weight with bice chair as the baseline. The screw is compact and rigid but it's heavier than the base line.

Syringe (Baseline)

Requirement ↓ \ Prototype →	Syringe (Baseline)	Bice Chair	Ratchet	Electromagnet	Screw
Material (durability / pen resistance)	0	- (0.0774 > 0.0211)	+ (0.01485 < 0.0211)	- (0.0582 > 0.0211)	- (0.032 > 0.0211)
Volume Closed (cm ³)	0	+ (75.9 < 176)	- (218.7 > 176)	+ (60.5 < 176)	+ (33.6 < 176)
Time & Steps (s / step)	0	- (51.08 / 4 > 47.08 / 5)	+ (17.33 / 1 < 47.08 / 5)	+ (17.95 / 2 < 47.08 / 5)	NA
Weight (g)	0	- (16.63 < 120.02)	- (29.31 < 120.02)	- (81.64 < 120.02)	- (36.28 < 120.02)

INTERPRETATION:

- ❖ Bice chair acts better on volume, but it is heavier and slower than the syringe in time and steps
- ❖ The ratchet and electromagnet have better durability, volume constraints and installation speed, given their compact nature.
- ❖ Screw is good under volume but fails the time evaluation

IV. Decision

[Video](#) testing metric of time and installation steps.

3 Key Design Decisions:

When we designed the syringe, we wanted the solution to be able to provide symmetric support to the screen. This ensures even wear on the laptop's hinges and reduces the chance of breaking the device. In the FEA simulations, results show the syringe solution causing a maximum displacement of 0.02mm when applying a force of 4.25N to the middle of the top of the screen (furthest point from both supports). This is less than many of our other designs such as the Bice chair and screw. Additionally, though not reflected in our pugh charts, a symmetrical design like this adds redundancy to the system, if one side breaks, the design can still function using the other side.

In the Pugh charts, the time of installation for the syringe appears to be very large, however, though the initial installation time is large, this design only needs to be installed once, and every time following the first would effectively take no time to set up. Since the syringe takes less time to set up than the beach chair (the other permanent design), it can be considered the fastest for setup. If we look into this a bit further, we can see that setting up the syringe once is equivalent to setting up the ratchet and electromagnet about 2.6 times. Considering that the average EngSci day contains about 4 lectures, even the total setup time for 1 day of use would be more for the temporary add-ons.

In the Pugh chart, the syringe didn't perform well in volume. Because the design is symmetric, it takes up twice the space of any other design. However, one thing to consider is the fact that upon installation, the part becomes a part of the laptop, and the overlapping volume would then be double counted. Realistically, in its closed state, the syringe is the only thing outside the laptop that needs its volume added to the amount of space taken up in a backpack.

6. Conclusion

Over the course of two months, our team dreamed, designed and built a hinged add-on (HAO) capable of improving stylus-based writing stability in 2-in-1 laptops for engineering science students. Through primary and secondary research, FEA, and iterative prototyping in CAD, we identified key performance needs: rigidity under pen pressure, volume, portability, and compatibility with the Lenovo Yoga i7. Five divergent concepts were generated, and comparative testing showed that the syringe-based mechanism offered the best balance of stability, smooth motion, manufacturability, and ease of use. FEA analysis further confirmed that material selection, especially adjusting components beyond ABS where required is critical for meeting the displacement limit of 0.775 mm under stylus loads. Overall, this design process produced a viable and user-centered hinge add-on concept that meaningfully improves the writing experience for engineering students while remaining compact, lightweight, and practical for everyday use.

7. Sources

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8. Appendix

Section A - Pairwise Comparison Matrix

Below is our pairwise comparison matrix which helped us compare each design against each other and form a better understanding of which final design we wanted.

	Material selection for durability and resistivity to pen pressure	Volume	Time and Steps	Weight	Total
Material selection for durability and resistivity to pen pressure	—	1	1	1	4
Volume	0	—	1	1	3
Time and Steps	0	0	—	1	2
Weight	0	0	0	—	1

Figure : Pairwise comparison matrix used to evaluate and rank the design requirements. Each requirement was compared against others to determine its relative importance, allowing the team to establish its weighted priority for evaluation.

Section B - Material Research

For materials, we decided to stick to plastic for initial prototyping. After finalizing our designs in Solidworks, we ended up exporting parts as STL's and 3D printing them out of PLA plastic. This is a common type of plastic for 3D printing that allows quick iteration but at the cost of quality. This plastic melts at relatively low temperatures and is commonly used for 3D printing to understand mechanical behaviours rather than final prototyping. Below are some examples of our prototyping:



As seen in each prototype, some layers of the print are falling off and separating, resulting in a part that buckles at low forces. This is also why our parts out of plastic can't be 3D printed, they must be made from a process that avoids layers and instead creates one complete part. An example could be injection molding but out of stronger materials.

After this realization, our group began researching commonly used and manufactured materials that provide good resistance to buckling. A material that we personally knew from 3D printing, matched this requirement with a high E value and resistance to bending. Below is the extract from the source we found that looked at highly used, high strength materials:



Not every kind of plastic is suitable for these tough jobs. But the ones that are can help streamline workflow, save money and complete projects more effectively than ever before.

Acrylonitrile Butadiene Styrene (ABS)

[ABS plastic sheet](#) is one of the most widely-used plastics on the market for its affordability and toughness. Other key features that make it ideal for heavy-duty plastic equipment include its incredible impact resistance and ease of machinability, requiring little power to be shaped and finished to the desired dimensions for an application. It also responds well to thermoforming, so it can easily be heated and bent into shape.

Common Applications of ABS Plastic Equipment

- Machine housings for factory equipment
- Vehicle instrument panels
- Storage cases on construction sites
- Prototyping and modeling
- Guards and covers for machines/equipment



We were particularly intrigued by the guards and covers as it implies high strength use and resistance to deformations. But upon FEA analysis, we discovered this E value was not enough as the material exceeds the buckling deformation limit. We still liked the unique benefits of ABS and so we looked into a heterogeneous composite material of ABS and a strong material. An ABS polycarbonate (ABS-PC) blend became the most promising candidate because it increases the effective Young's modulus and tensile strength compared with pure ABS, while also dramatically improving toughness and heat resistance. Below is what we looked at:

Polycarbonate (PC)

Polycarbonate is the toughest plastic out there, at more than 20 times stronger than acrylic and 200 times stronger than glass. It has great clarity, so it's frequently used in the types of plastic equipment that require see-through applications. Polycarbonate also has a great balance of engineering properties that give it the ability to operate in a broad range of temperatures, resist impact and maintain dimensional stability. It's also a thermoplastic, so it can be reheated and remolded into new shapes as needed.

Common Applications of HDPE Plastic Equipment

- Industrial and commercial glazing (windows)
- Transparent machine guards
- Heavy machinery windows and windshields
- Face shields and sight glasses
- Semiconductor components

After more connections failed the deformation requirement, we decided to switch paths and look at materials which require more effort to manufacture, but significantly increase the strength. After looking at industry used metals and the young's modulus, we settled on ferritic stainless steel. By comparing Young's modulus values across common engineering metals, ferritic stainless steel stood out as the most appropriate choice. Its high stiffness, good resistance to deformation under bending, and widespread use in load-bearing laptop components made it the best balance between strength, durability, and practical manufacturability.

metals.com

GRADES OF FERRITIC STAINLESS STEEL AND USES INCLUDE:

- **409 stainless steel:** Automotive exhaust systems, electrical kiosks and distribution boxes
- **430 stainless steel:** Automotive trim, chimney liners, dish washer linings, fasteners, refrigerator cabinet panels and stove trim rings.
- **430LI stainless steel:** Used for appliances, architectural screens, cooking utensils, fasteners, food processing and interior trim.
- **434 stainless steel:** Architectural applications, automotive trim, dishwashers, downspouts and gutters, gas burners, heating units, oil refinery equipment, restaurant equipment, roofing and siding.
- **439 stainless steel:** Automotive exhaust manifolds, hot water tank components and residential furnaces, as well as in the chemical processing, petroleum refining and power generation industries.
- **442 stainless steel:** Food processing equipment, heavy gauge welded components, pharmaceutical processing equipment and surgical implants.
- **444 stainless steel:** Cladding and roofing in marine environments, food processing equipment, hot water heaters and stainless steel tubing for heat exchangers.
- **446 stainless steel:** Boiler components, burners, flame rods, injector nozzles for molten compounds and thermocouple protectors.



MAIN PROPERTIES OF FERRITIC STAINLESS STEEL

While heat treatment won't harden or strengthen ferritic steels, they can be cold-worked and softened through an annealing process, which involves heating the metal and then slowly cooling it. Though they're neither as strong nor as resistant to corrosion as austenitic steels, ferritic grades normally have more beneficial mechanical properties. They tend to be easy to weld, though some ferritic grades have limitations, restricting use of these steels to thinner gauges.

Because they don't contain nickel and have lower amounts of chromium, standard ferritic stainless steels normally cost less than austenitic grades. The most widely used ferritic grade of stainless steel is SS430. With lower chromium content, SS409 is less corrosion-resistant. In contrast, SS439 resists corrosion better and is more stable in high temperatures. Depending on the alloy, ferritic stainless steels exhibit several beneficial properties.

CORROSION RESISTANCE

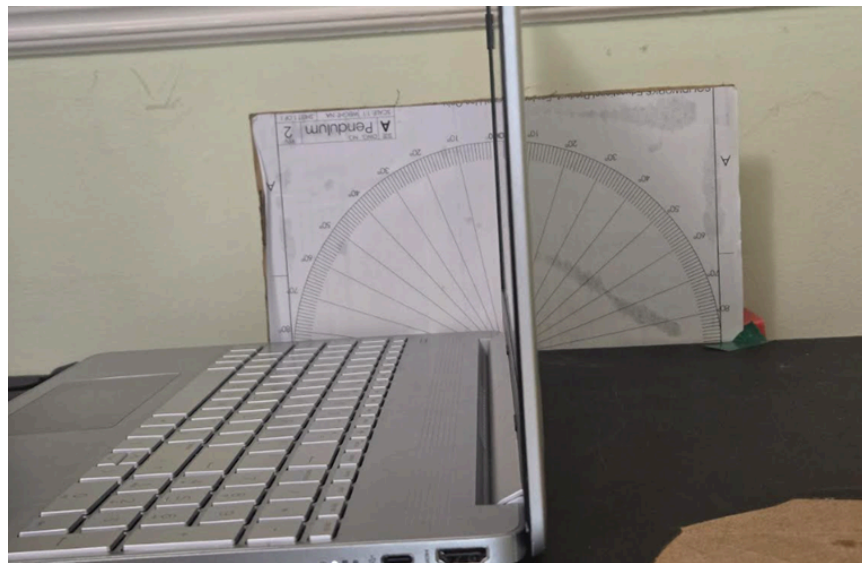
After these changes, the model met the deformation standard.

Section C - Acceptable Displacement Calculations

Part 1: Acceptable movement angle

From Design Brief Appendix C:

By reusing materials and skills from the PHY180 pendulum project, we were able to roughly estimate what is the range of oscillation that is noticeable by the human eye. Screen bounce is caused by a very minimal force applied laterally that due to the hinges lack of resistance, begins to oscillate. Even small angles such as 1 degree are noticeable by eye as it is a periodic motion that lasts a few seconds. As a result, we conducted a test after difficulty finding sources that investigate noticeable oscillation range of a screen.

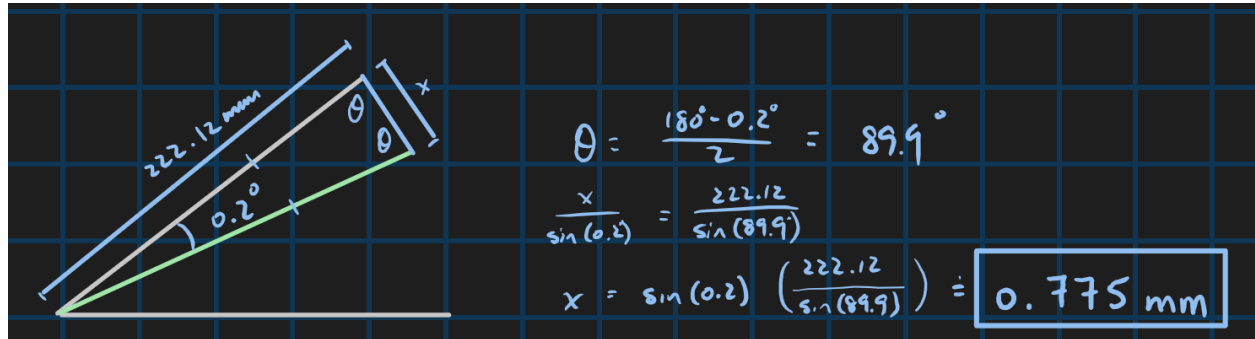


Our Configuration consisted of the laptop screen set to 90 degrees to the enlarged protractor made in SolidWorks and a video orthogonal to the laptop screen to measure the angle of oscillations. Gradually we

applied less force until from the distance of a typical desk, you would not be able to notice the oscillation. This gave us the oscillation of ± 0.2 degrees of what is not noticeable by eye.

This data to be more accurate must be tested with a more precise configuration with lower uncertainties as well as over a larger range of two in one laptop. Every brand and product is built differently, and the hinges react differently, thus getting an average between the differences would be valuable information.

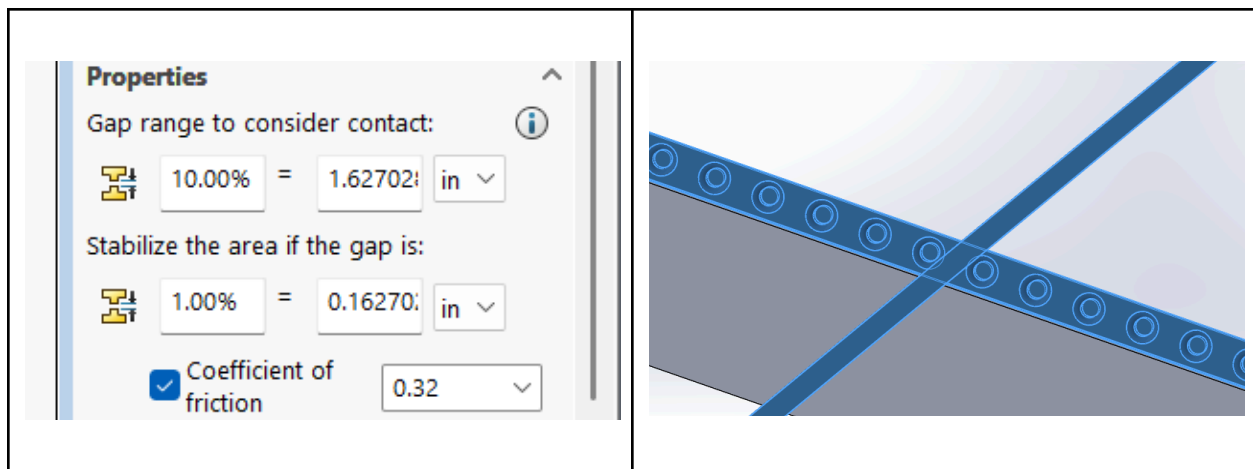
Part 2: Max displacement calculations



Using our primary research from the design report, we found that from our group the minimum angle of visual displacement that is noticeable to be ± 0.10 degrees. By considering the full length of the laptop (222.12mm), we found that the vertical displacement from the top to the 0.2 degree lines as shown in the figure above is 0.775mm in CAD. We decided to analyze vertical displacement because with Simulations, the vertical displacement is easy to find.

Part 3: FEA Research

Using non bonded type for FEA analysis, using average of smallest and max coefficient of friction for assumed material: ABS, it ranged from 0.18 to 0.42. This is based on material but it primarily fits in this range for 3D printed ABS. To find a balance, we took the average between the two getting a μ of 0.32 for FEA:



Simulation Input	Contact point (Example is at 20 degrees)
------------------	--

<https://www.plasticsintl.com/products/abs> - Source used

Part 4: Writing forces

Two secondary sources were evaluated using the CRAAP test and the results were compared. We took the highest value between the test as based on how it performs for the max, it implies it can survive the smaller forces.

Study 1: The sources provided describe a study on pen force during handwriting using a specialized experimental setup, not modern devices like iPads. The experiment utilized a large-size writing tablet (Calcomp 9000) and a laboratory-made writing stylus equipped with a strain-gauge force transducer

Condition	Average APF	Converted Force
C- (Counter-clockwise circling)	115.3g	$\approx 1.13 \text{ N}$
F (Straight finger movements)	113.5g	$\approx 1.11 \text{ N}$
H (Cursive handwriting)	109.7g	$\approx 1.08 \text{ N}$
C+ (Clockwise circling)	108.9g	$\approx 1.07 \text{ N}$
W (Straight wrist movements)	87.5g	$\approx 0.86 \text{ N}$
S (Scribbling)	83.6g	$\approx 0.82 \text{ N}$

<https://link.springer.com/article/10.1007/BF00203451> - Source used

Formula To Convert:

$$\text{Force (Newtons)} = \frac{\text{Force grams}}{1000} \cdot 9.80665 \frac{m}{s^2}$$

Study 2

Measurement index	Writing pressure task		Writing speed task	
	Strong	Weak	Fast	Careful
Pressure				
Pen pressure (g)	433 ± 92	83 ± 33**	222 ± 74	200 ± 63*
Writing surface pressure (g)	506 ± 114	83 ± 40**	248 ± 87	223 ± 76*
Hand pressure (g)	787 ± 259	350 ± 150**	432 ± 179	451 ± 176
Forearm pressure (g)	1,881 ± 730	1,888 ± 726	1,747 ± 639	1,815 ± 761
Degree of variation				
Pen pressure (ΔPi)	5,050 ± 830	4,060 ± 330**	5,040 ± 740	4,170 ± 450**
Writing surface pressure (ΔPi)	10,760 ± 3,100	9,220 ± 700**	10,690 ± 1,420	9,460 ± 1,200**
Writing speed (mm/s)	11.49 ± 0.31	11.70 ± 0.32**	21.85 ± 1.16	7.78 ± 0.12**
Range of motion				
Palmar flexion and dorsiflexion (degrees)	9.2 ± 2.5	7.7 ± 2.2**	8.1 ± 2.3	8.1 ± 2.5
Radial and ulnar flexion (degrees)	12.6 ± 3.3	11.0 ± 2.3**	11.3 ± 2.8	11.4 ± 2.8
Index finger PIP flexion (degrees)	19.0 ± 7.0	18.4 ± 6.0	18.7 ± 6.9	19.7 ± 6.2
Area of overlap (pixels)	435 ± 601	650 ± 712*	1,177 ± 1,202	229 ± 353**

Pen Pressure Source 2

https://www.researchgate.net/figure/A-comparison-of-the-handwriting-performance-grip-strength-and-pen-pressure-measures_tbl1_337482048 - Source

This figure from source [3] specifically the implication of the type of task and writing speed with respect to writing pressure. This can further be understood by comparing the numerical data with the source extract below which initially aided us in understanding that writing pressure has lots of variables and there is no simple average. Lots of factors such as what the task is, how fast you write, grip, posture and age must be considered for finding a specific pressure that causes the hinge to move.

Using source two we get a max force of **4.246236** Newtons that will be used in the FEA analysis.

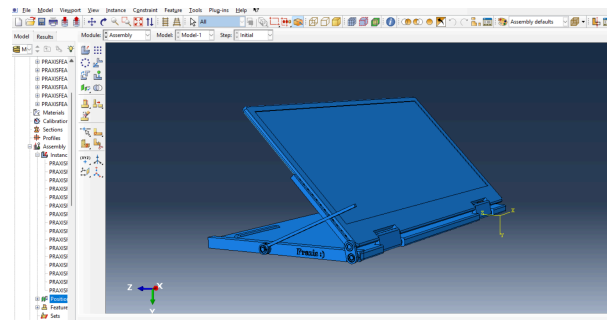
Section D- FEA Process

Step	Photo

Step 1 – Test Software

Abaqus Solidworks - Attempt 1

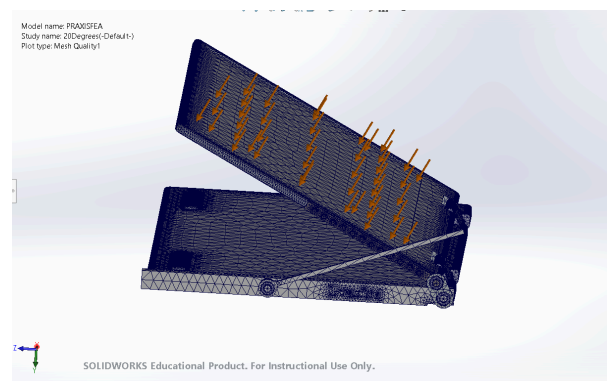
Initial attempt at testing FEA, difficulty lied in moving parts from one software to another, particularly in recreating the assembly with mates.



Step 2 - Solidworks Simulate Mesh

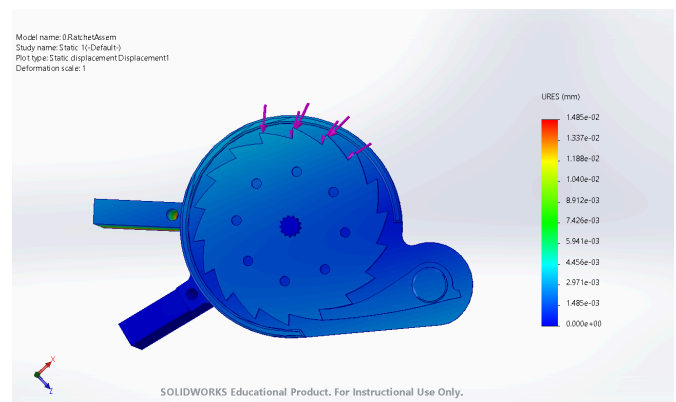
Mesh and apply fixtures as well as external loads.

The polygons represent the mesh of the laptop and bice chair at 20 degrees. The goal is to have a fine mesh for more accurate results. The orange arrows represent the applied force.



Step 3 - Solidworks Simulate Results

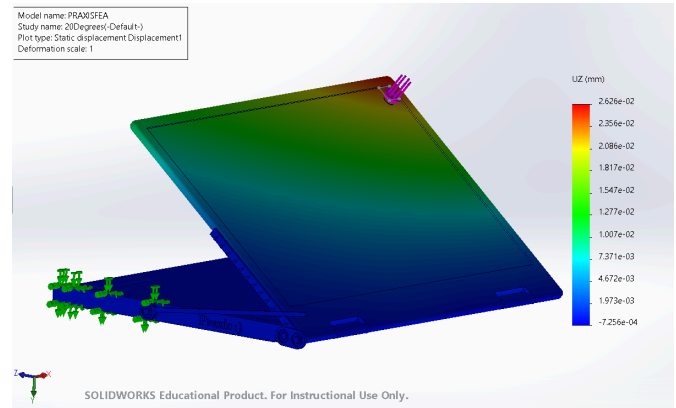
By stimulating the force applied on the screen just like a person does to the laptop, we are able to see both the deformation of the material and the stress. Similar to the bridge project, from here we are able to iterate and decide upon a better material to move forward with.



Step 4 - Solidworks Simulate Results Point Force

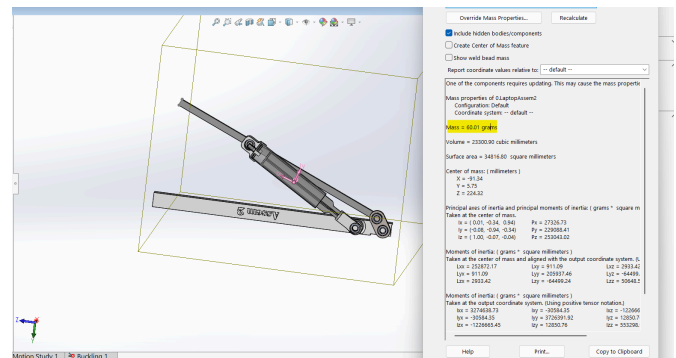
Case 2 : Point Source Far Away

With a further point source as seen in the figure on the right, at larger distances the screen itself displaces a maximum of 0.07mm and the Bice Chair itself resists movement. This shows that a tougher material compared to steel is required, since ABS cant withstand the force.



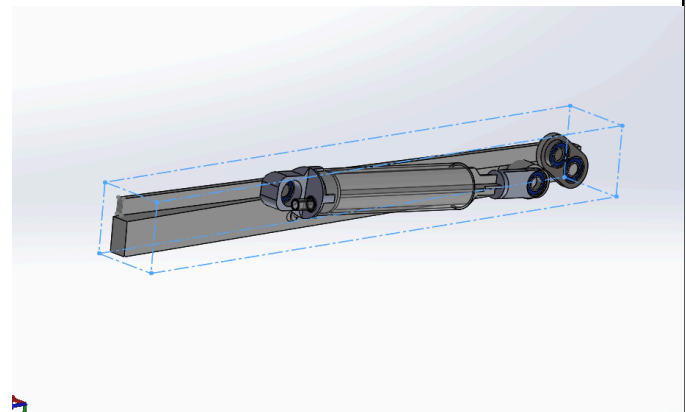
Step 5 - Mass Analysis

By determining the mass to be either ABS, ABS PC (Polycarbonate) or ferritic stainless steel we can find the exact mass of the mechanism. The smallest mass relates the most to our goal of portability and easy add on.



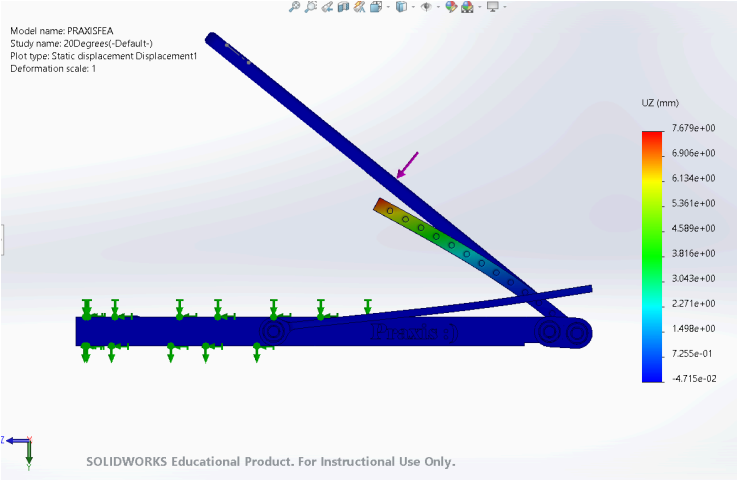
Step 6 - Volume Analysis

After the primary analysis is completed, the volume when closed can be calculated. The smallest volume and dimensions demonstrates a prototype that relates the most to our goal of portability. A rectangle box of best fit was used to calculate the smallest occupational volume.



In addition to the syringe, we considered the material iteration for the Bice chair. Each row depicts a new material tested and the displacement. For the Bice Chair, with ABS (Row 1) the deformation was 10 times larger than our requirements of 0.775 and even when we changed to ABS Polycarbonate (Row 2) the deformation only changed by nearly 1mm to 6.37mm. This made us realize that plastics are not the required material for such

a solution and thus we looked at common metals such as steel and found ferritic stainless steel, a commonly used metal that provides large resistance to bending/deformation (Row 3). The results were within the bounds of the requirements as shown below:



Model name: PRA20SFEA
Study name: 200Degrees(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 1

UZ (mm)

7.679e+00
6.906e+00
6.134e+00
5.361e+00
4.589e+00
3.816e+00
3.043e+00
2.271e+00
1.498e+00
7.255e-01
-4.715e-02

SOLIDWORKS Educational Product. For Instructional Use Only.

Minimum Z Axis Displacement: 0 mm

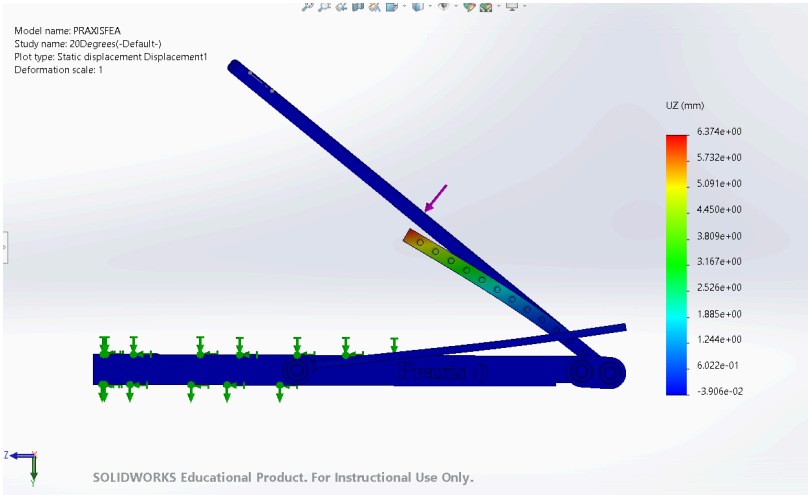
Maximum Z Axis Displacement: 7.697 mm

Property	Value	Units
Elastic Modulus	2000000000	N/m^2
Poisson's Ratio	0.394	N/A
Shear Modulus	318900000	N/m^2
Mass Density	1020	kg/m^3
Tensile Strength	30000000	N/m^2
Compressive Strength		N/m^2
Yield Strength		N/m^2
Thermal Expansion Coefficient		/K
Thermal Conductivity	0.2256	W/(m·K)

Material Property: ABS

Key characteristic:

E: 2000000000 KPA



Model name: PRA20SFEA
Study name: 200Degrees(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 1

UZ (mm)

6.374e+00
5.732e+00
5.091e+00
4.450e+00
3.809e+00
3.167e+00
2.526e+00
1.885e+00
1.244e+00
6.022e-01
-3.906e-02

SOLIDWORKS Educational Product. For Instructional Use Only.

Minimum Z Axis Displacement: 0 mm

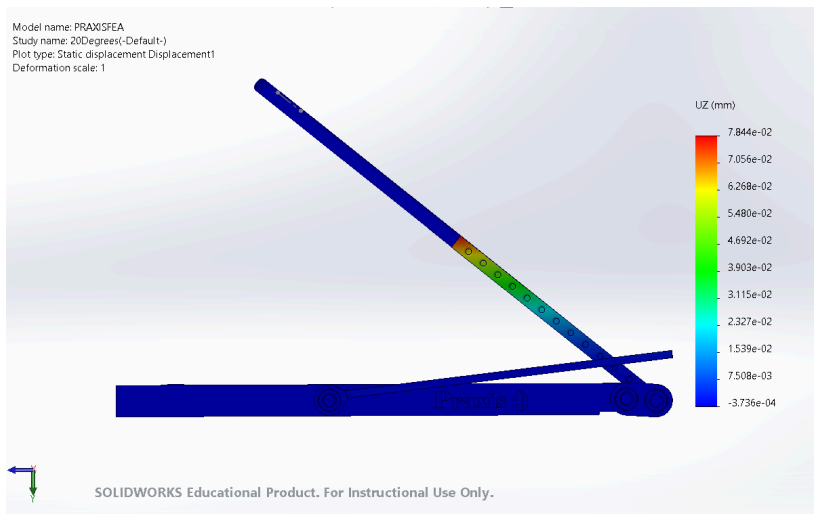
Maximum Z Axis Displacement: 6.37 mm

Property	Value	Units
Elastic Modulus	2410000000	N/m^2
Poisson's Ratio	0.3897	N/A
Shear Modulus	862200000	N/m^2
Mass Density	1070	kg/m^3
Tensile Strength	40000000	N/m^2
Compressive Strength		N/m^2
Yield Strength		N/m^2
Thermal Expansion Coefficient		/K
Thermal Conductivity	0.2618	W/(m·K)
Specific Heat	1900	J/(kg·K)

Material Property: ABS

Key characteristic:

E: 2410000000 KPA



Property	Value	Units
Elastic Modulus	2e+11	N/m ²
Poisson's Ratio	0.28	N/A
Shear Modulus	7.7e+10	N/m ²
Mass Density	7800	kg/m ³
Tensile Strength	513613000	N/m ²
Compressive Strength		N/m ²
Yield Strength	172339000	N/m ²
Thermal Expansion Coefficient	1.1e-05	/K
Thermal Conductivity	18	W/(m-K)

Material Property: Stainless Steel
Ferritic

Key characteristic:

$$E: 2.0 \times 10^{11} \text{ KPa}$$

Minimum Z Axis Displacement: 0 mm
Maximum Z Axis Displacement: 0.07844mm

For using FEA with Solidworks, we used source 13 to help get a better understanding of static conditions. Here are some aspects that proved valuable when we got stuck (General process):

4. Apply Loads

Specify any faces, edges, and vertices to apply external loading. Common loads include forces, pressure, and torques. Loads can be uniform or follow non-uniform distribution.

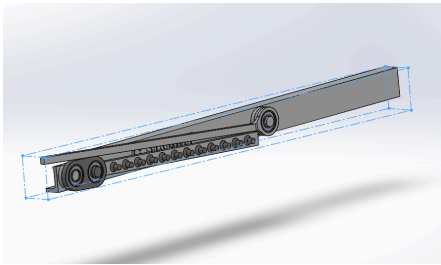
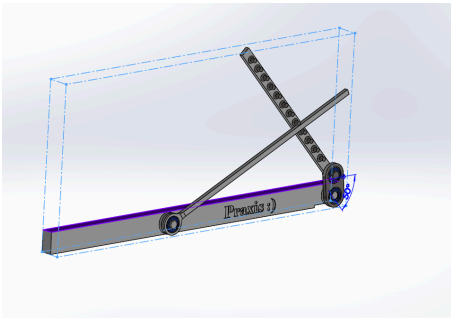
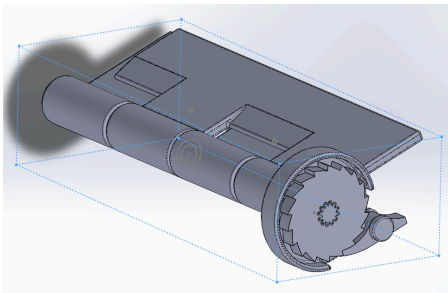
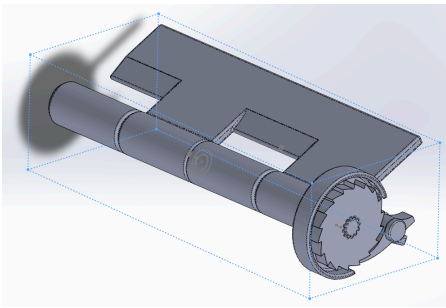
500N External force applied to side face

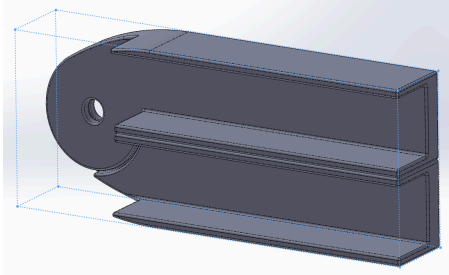
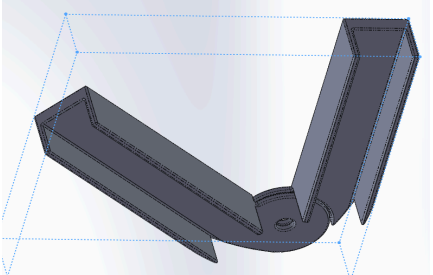
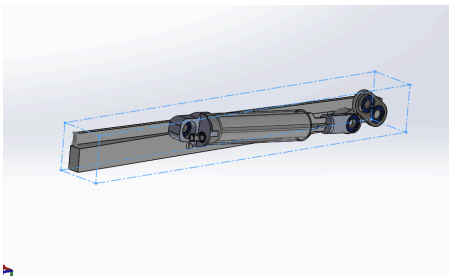
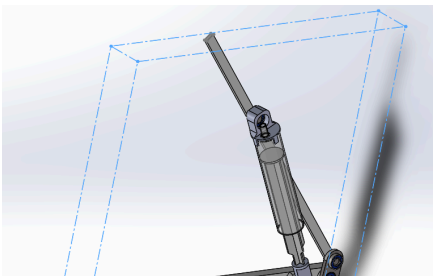
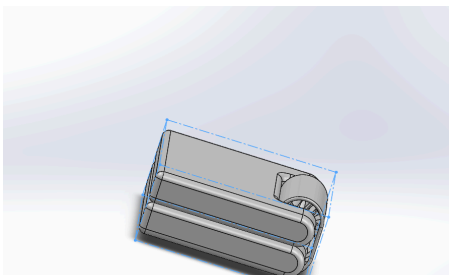
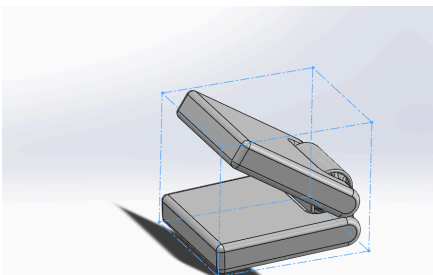
5. Mesh the Model

Meshing divides the model into smaller, more manageable triangular elements to analyze, hence the term finite element analysis (FEA). The triangular meshes can be coarsely sized or set much finer, this helps find a balance between the time required to run the study and the accuracy of the results.

Section E - Volume Calculations

When considering volume, we found the minimum volume the solution occupies given by the shape of a rectangle. Using solidworks, this was easy as all we needed to do was define boundary space and analyze the properties of it to get the resulting volume. Here are the results:

Design	Closed Volume (mm)	Max (50°) Volume (mm)
Bice Chair	22.61mm x 14.06mm x 238.69mm 	14.06mm x 105.37mm x 224.01mm 
Ratchet *Note - Assuming hinge goes half length of the laptop*	127.24mm x 52.07mm x 33.01mm  As closed as possible, near 7 degrees	127.2mm x 50.42mm x 36.64mm 
Screw	80mm x 30mm x 14mm	80mm x 78.1mm x 14mm

		
Syringe Note - two of these components	28.6mm x 27.3mm x 225.43mm  As close to closed as possible	28.62mm x 184.9mm x 226.33mm 
Electromagnet	54.84mm x 24.52mm x 45mm 	58.0mm x 44.57mm x 45mm 

Initially we considered the closed and max volume but upon further consideration, we realized max volume at 50 degrees provides no value. The initial idea was to differentiate designs such as the Bice Chair since the arm at a maximum angle of 50 degrees sticks out into the user's oath of writing therefore limiting the design to one side. But we figured that it's best to talk about it separately than have a metric that doesn't directly show this problem.

Section F - Minimum and Maximum Angles Primary

Discussed in the background section 1.1. Where flexibility meets fragility, this was a mix of primary and secondary research to determine a general range of angles based on common first-year engineering science student classrooms, including McLennan and Sandford Flemming. The results proved the need for adjustable hinges

i) Find Average Height

$$H_{avg} = \frac{\text{Height 1} + \text{Height 2}}{2}$$

$$H_{avg} = \frac{1533 + 1694}{2} \text{ mm}$$

$$H_{avg} = 1613.5 * 0.1 \text{ mm}$$

$$H_{avg} = 161.35 \text{ cm}$$

Convert to cm as we did all our measurements in cm, easier to convert one to cm than to convert the other to inches

This calculation gives us the average height between males and females with boots. We decided to include boots, as in class we also have shoes on, and the difference between with and without is not negligible, as per the tables on the right. While military shoes are thicker, the difference compared to regular shoes is not too big of a difference.

Male Height with Boots 5th Percentile:

Hand Length	179	186	192	199	213	193	10
Head Breadth	144	149	153	157	163	153	6
Head Length	187	195	200	204	208	199	7
Height of Chest (Chest Height)	1207	1250	1280	1314	1370	1284	52
Height of Hip Circumference (Buttock Height)	821	859	884	908	976	887	46
Hip Circumference (Buttock Circumference)	924	982	1043	1087	1157	1042	75
Knee Height, Sitting	513	534	552	571	600	553	27
Maximum Hip Breadth, Sitting	325	343	364	385	413	365	28
Popliteal Height	426	440	455	470	508	458	23
Radiale-Styleon Length	242	253	261	271	292	262	16
Stature Without Boots	1648	1710	1752	1784	1860	1749	63
Stature With Boots	1694	1752	1794	1821	1891	1789	61
Trigono-Top of Head	121	128	132	135	140	131	6
Waist Circumference (Omphalion)	750	826	915	983	1043	906	104
Waist Height (Omphalion)	976	1016	1046	1075	1132	1047	48
Weight (kg)	61	73	83	90	103	82	13

* mm unless noted

Female Height with Boots 5th Percentile:

Height of Hip Circumference (Buttock Height)	767	814	828	859	885	831	37
Hip Circumference (Buttock Circumference)	933	969	1020	1070	1104	1019	70
Knee Height, Sitting	474	498	513	531	547	514	23
Maximum Hip Breadth, Sitting	337	358	373	394	425	377	30
Popliteal Height	383	412	428	436	456	424	21
Radiale-Styleon Length	226	235	242	254	260	243	12
Stature Without Boots	1502	1586	1619	1670	1716	1619	66
Stature With Boots	1533	1620	1653	1696	1753	1654	65
Trigono-Top of Head	118	124	127	130	142	127	7
Waist Circumference (Omphalion)	727	759	815	907	986	838	90
Waist Height (Omphalion)	899	935	972	1005	1026	968	44
Weight (kg)	55	59	62	70	83	65	9

* mm unless noted

*Units are in mm

Using the 5th percentile rather than 25th percentile despite the large difference to the 25th is significant, because this way we can account for a larger percentage of our class using this assumption.

ii) Find Ratio Buttocks to Eye Level

$$Dist = 0.520h - 0.065h$$

$$Dist = 0.455h$$

Sub in calculated height

$$Dist = 0.455(161.35cm)$$

$$Dist = 73.414 cm$$

The report uses a ratio of height to body components. By finding the average height and multiplying by that ratio I found height buttocks to eye level. Later on, I found another source in which my numbers differed by around 0.5cm [12].

iii) Find Angle of Depression

Using these dimensions, we went to two primary classes as engineering science students spend a lot of time using (SF and MP).

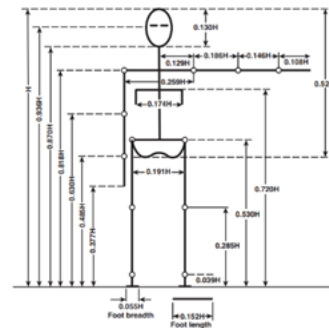
For McLennan room 203, the bottom of the chair to the top of the desk measured 26cm whilst the distance from the chest to the start of the desk was 31 cm.

For SF 1105, the height from the bottom of the chair to the top of the desk measured 22 cm whilst the distance from the chest to the start of the desk was 31cm.

Refer to Figure for dimensions

Ratio Of Height Of A Human

Figure 39.2: Correlation coefficient.



Body segment lengths expressed as a fraction of body height H

Figure 39.3: Link-length proportion. Source: Drillis and Contini 1966.

Further Study Confirming This Ratio: By looking at the 5th percentile, the height in cm is 72.66 cm

Centimeters	Mean	Std Dev	Maximum	Minimum	Percentiles	Inches
72.85	28.68	3.31	84.00	63.00	1 st	28.68
72.85	28.68	3.31	84.00	63.00	2 nd	28.68
72.85	28.68	3.31	84.00	63.00	3 rd	28.68
72.85	28.68	3.31	84.00	63.00	5 th	28.68
72.85	28.68	3.31	84.00	63.00	10 th	28.68
72.85	28.68	3.31	84.00	63.00	15 th	28.68
72.85	28.68	3.31	84.00	63.00	20 th	28.68
72.85	28.68	3.31	84.00	63.00	25 th	28.68
72.85	28.68	3.31	84.00	63.00	30 th	28.68
72.85	28.68	3.31	84.00	63.00	35 th	28.68
72.85	28.68	3.31	84.00	63.00	40 th	28.68
72.85	28.68	3.31	84.00	63.00	45 th	28.68
72.85	28.68	3.31	84.00	63.00	50 th	28.68

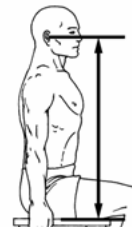
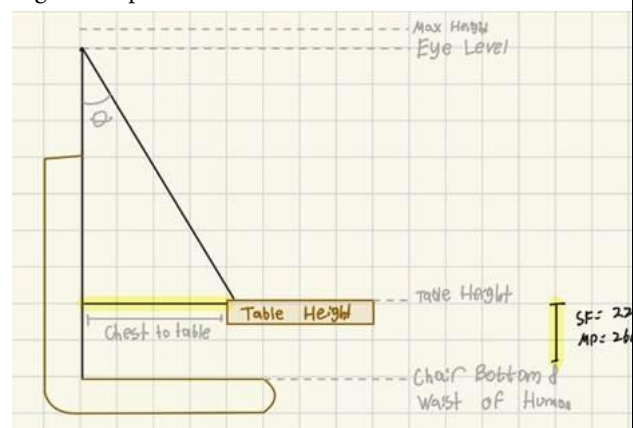


Diagram of the configuration used to calculate the angle of depression:



Using these calculations, we created a right-angle triangle by subtracting average height by the chair to table height. Finally we can find the maximum angle of depression from the student's eye to table that they would have to look at.

$$\theta = \arctan \frac{\text{Chest to table distance}}{\text{Avg Height} - \text{Chair height}}$$

For MP:

$$\theta = \arctan \frac{31}{73.414 - 26}$$

$$\theta = 33.14^\circ$$

For SF:

$$\theta = \arctan \frac{18}{73.414 - 22}$$

$$\theta = 19.3^\circ$$

These two angles reflect further research we found on angle relation to comfort whilst taking notes [5]-[3] as 34 degrees is what's described as the average comfort angle but in general the hinge should focus on $20^\circ - 50^\circ$.

The highlighted aspects are the dimensions mentioned that we measured.

Additionally, below you will find some photos of these classrooms and the difference in configuration:



Figure A Depicts Sanford Flemmings front row seats. Notably it has cushioning and a unique arm rest

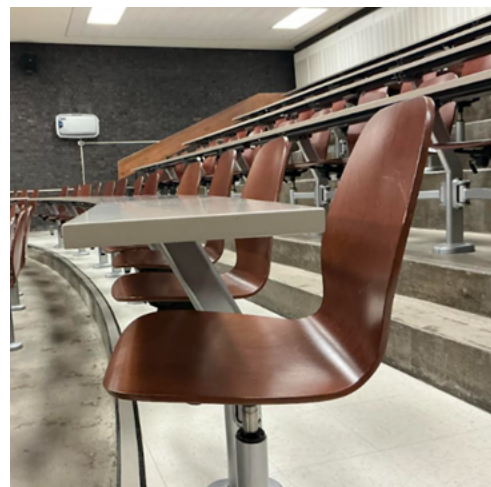


Figure B depicts McLennan's back row seats. Notably it's a thin chair that can rotate around but as a result is very close to the table.

Discussed in the background, these are the calculations for our minimum viewing angle given the mean height of soldiers, a good way of finding an abundance of data about their height and proportions. Using these proportions, we found the average screen viewing angle for classroom SF1105 and MP 203. A clear difference emerged of viewing angles by roughly 10 degrees, and this revealed the need for a hinge that not only is adjustable, but rigid based on user needs.

Section G - Design Brief

Design Brief

1 Introduction

Convertible laptops, a modern alternative to handwritten notes, rely heavily on a stable surface to work optimally. The ability to transform, however, often comes at the cost of hinge rigidity. The current laptop design cannot resist the downward and lateral pressure applied during stylus use, leading to screen wobble and user discomfort. This negatively affects tasks that require precision and control, thereby limiting the device's usability in tablet mode. While convertible laptops promise flexibility and convenience, users face a trade-off between flexibility and functionality.

1.1 Where Flexibility Meets Fragility

For such a revolutionary product first introduced in the 1990s [14], offering the benefits of a laptop's speed and simplicity, whilst also including the efficiency of digital notetaking on a tablet, the idea is perfect, but not the implementation. Particularly for Engineering Science (EngSci) students who already operate under tight schedules, this duality promises simplicity and productivity. This would benefit at least 42% of students [11] at the University of Toronto (UofT) who balance between paper notes and a laptop into one portable device. Currently, convertible laptop hinges do not provide sufficient stability for effective stylus use in tablet mode, limiting the device's usability for tasks that require precision. Although this seems minor, even small ergonomic discomforts can lead to a lack of efficiency over time.

Upon analyzing current first-year EngSci lecture rooms, by finding the dimensions of the classroom chairs of McLennan and SF1105 along with the mean distance from lower waist to eye level of US

soldiers [4]-[12] of similar age (Average of 24), we discovered that the natural angle of depression fluctuates from lecture hall to lecture hall. As a result, the need for an adjustable hinge for notetaking is essential, primarily between 19° - 33° from our analysis of our classroom [see [Appendix A](#)]. But with a lack of resistance, students often refer to keeping their tablets flat due to a lack of solution for the duality of strong but adjustable hinges, as per our interviews with fellow UofT students [[Appendix C](#)]

2 Stakeholder Values and Interpretations

Understanding what matters to stakeholders helps justify how we frame the opportunity and set the basis for Needs, Goals, and objectives (NGO's).

2.1 Primary Stakeholders (PS)

This design is aimed towards first-year Eng Sci students who use 2-in-1 laptops for notetaking, sketching, and other stylus-based activities. They are directly affected by the shortcomings of the current hinge system. Observations suggest students often voice concerns about wobbly screens or awkward sitting positions. They are considered PS because screen instability impacts their productivity, comfort, and academic performance. They often **switch** between modes, some up to 10 times a day ([Appendix C](#)), making ease of transformation critical. Portability is key, as students carry their laptops around campus daily. Any add-on device should not increase bulk. **Aesthetic** appeal also matters, as the design should look professional and appropriate for an academic environment, rather than an afterthought. We interpret their needs as **valuing precision, stability, and an ergonomic experience.**

2.2 Secondary Stakeholders (SS)

While the immediate stakeholders are first-year EngSci students, the potential solution could also benefit a wider audience. This includes students in other faculties, educators, and professionals in a wide range of industries. When researching designs, it may be beneficial to consider design adaptability in case modifications are required. These stakeholders value similar qualities as PS ([Appendix C](#)). They prioritize solutions that are **effective and reliable in maintaining stable stylus** interaction during use. Their inclusion is justified by the broader applicability of the design opportunity.

3 Design Team Values and Interpretations

When addressing the opportunity, the design team should prioritize core values that align with stakeholder expectations as well as bring something new to the table. While EngSci's students may seek greater stability or comfort, we understand these as an association with ergonomic support, intuitive use, and dependability. Broader considerations, like accessibility and inclusivity, though not always explicitly requested, are essential to ensure that the solution serves a diverse user base. The final solution should be inclusive (usable by individuals of a range of physical abilities), durable (a design that maximizes its lifespan through robust construction), interoperable (compatible access to various devices), and ergonomically safe.

4 Engineering Needs, Goals, and Objectives (NGOs)

Need:

To address the lack of screen stability in convertible laptops during stylus use in tablet mode, to meet and exceed students' expectations for a consistent, comfortable, and reliable writing experience. (see [Appendix A](#))

Goal 1: Range of Motion

Design a durable, repairable hinge for a 2-in-1 laptop that can maintain rigidity but enable smooth motion across a range of 0° - 360° . Everyday writing and external forces should be withstood to minimize natural screen vibrations. This ensures a long-lasting product and premium user experience.

Objective 1.1: Hinge should resist deformation or flexing of at least the typical maximum writing force on tables of 433 ± 92 grams of applied force (See Appendix C for more information). While writing pressure typically depends on speed [1] and specific pencil grip, ensuring our product is rigid at any set angles whilst allowing users to easily switch between 0° - 360° is a priority.



Figure 1: Three rendered configurations of a typical HP two in one laptop, illustrating the range of hinge motion from fully closed to standard laptop and writing positions, created using SolidWorks Visualize. [13]

Objective 1.2 Enhance screen stability by limiting visually perceived screen vibrations through screen design considerations like stiffness, damping and center of mass. The screen should be limited to an angular movement within $\pm 0.2^\circ$ (See Appendix D) to decrease the effects of natural oscillations by the screen.

Objective 1.3: Upon review of customer feedback where some state, “laptop hinges are prone to breaking with normal usage” [9], its clear hinge quality is often overlooked within the current market.

As a result, our product must adhere to similar hinge standards, such as the BHMA A156.1 standard [10], in which lightweight door hinges must pass 350,000 oscillations to become certified. This ensures customers a product that is reliable and capable of withstanding four years of labs, calculus homework and all-nighters.

Objective 1.4: The solution must be symmetrical across the screen of the laptop, such that use of the device does not cause asymmetrical loading on the existing hinges.

Goal 2: Ergonomics

Optimizing user posture and position by allowing the tablet to rest at an ergonomically ideal angle during lecture notetaking. This will be achieved through a hinge design that offers both adjustability and stability when fixed, which enables students to efficiently take notes without discomfort or fatigue.

Objective 2.1: Allow a range of tablet angles relative to the table from a range of 20° to 50° [7] which we proved applicable in first year engineering science classrooms. By maintaining proper arm and wrist positioning, users can achieve ergonomic neutrality in which minimal force is required to rest in a writing position. A widely recognized technique is the Palmer Method [8] which emphasizes natural arm movement in which your forearm, wrist, and hand move as coordinate units. By resting forearms at least above 0 degrees from the tabletop, neutral wrist movement, and bending whilst ensuring stylus motion comes from forearms rather than finger movement limits soreness for students.

Objective 2.2: Reduce the amount of body posture deviation, such as leaning forward or wrist overextension, by keeping the upper body upright and joints (hips, knees, ankles) within the recommended ergonomic neutral range (approximately $90^\circ \pm 10^\circ$) [6] as observed in figure 2. As a result of these changes, users can get more out of the product by significantly decreasing discomfort due to excessive neck and shoulder movements [5].



Figure 2: Depicts the common postures by students in lectures, in which the ideal posture for comfort is the individual on the left. View Appendix C, section D for a real life example of posture.

Goal 3: Portability

The solution must be portable for EngSci students traveling between classes while maintaining symmetry to prevent unbalanced stress due to asymmetry. The symmetrical aspect will also aid in contributing to a sense of harmony in the final solution, enhancing both portability and aesthetic appeal.

Relevant research can be read in [Appendix B](#).

Objective 3.1: The solution must be contained within the bounding box in Figure 3. (Note that all dimensions are in cm).

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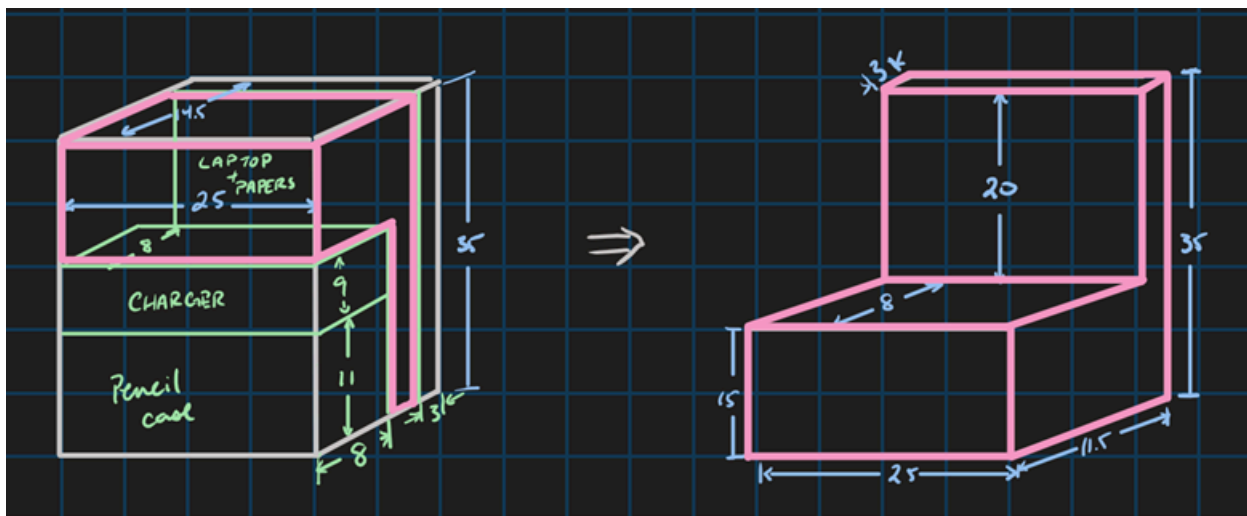


Figure 3: Calculated bounding-box for possible solutions. Note that this is not a suggestion for the shape, but rather a shape to allow choice between a flat and/or boxy form factor.

Figure 3: Calculated bounding-box for possible solutions. Note that this is not a suggestion for the shape, but rather a shape to allow choice between a flat and/or boxy form factor.

Objective 3.2: The solution must not weigh more than 300g. The metrics of the maximum weight is based on the maximum recommended back carrying capacity of a 5th percentile female aged 18. The available weight was determined by summing the weight of an empty backpack, a pencil case, a charger, a phone, and a filled water bottle.

5 Reference Designs

Current designs for hinges compromise mechanical rigidity and the intended “laptop-tablet The slant board (Figure 3) commonly used to address writing posture by the stakeholder. Though its bulk and size make it unsuitable for portability or use in classes.

This reveals a trade-off between stability and portability, suggesting that the potential design solution should integrate the angle adjustability of a slant board in a more compact form, functionality in favor of thinness and cost-effectiveness. Reference studies show typical hinge torque tolerances around $\approx 0.6 \text{ N} \cdot \text{m}$, whereas stylus input can generate higher localized forces, exceeding hinge design specifications.

Similarly, the drafting desk mechanism shown in Figure 4 is a robust, adjustable-angle platform suitable for tasks that involve precision. It highlights the potential for rigid support at various angles, and this idea should be adapted to be integrated within a laptop form factor.

6 Conclusion

This design brief focuses on the need for improved screen stability in convertible laptops during stylus use. EngSci students want a consistent, reliable, and comfortable experience that supports their academic work. Through interviews, journals, and research papers, various goals and objectives were defined to guide the design team in producing a well-rounded device that emphasizes functional hinge performance, ergonomic support, portability, and aesthetics. However, comfort and usability are subjective topics. While research provided guidance on effective ergonomics practices that the future device should follow, these factors are often influenced by individual writing styles, posture habits, and personal preferences. Thus, ergonomic principles and technical benchmarks will guide the future design, though the final solution must remain adaptable to diverse user behaviors and expectations.

Section H - Source Extracts

This figure from source [3] specifically the implication of the type of task and writing speed with respect to writing pressure. This can further be understood by comparing the numerical data with the source extract below which initially aided us in understanding that writing pressure has lots of variables and there is no simple average. Lots of factors such as what the task is, how fast you write, grip, posture and age must be considered for finding a specific pressure that causes the hinge to move.

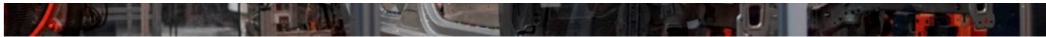
Measurement index	Writing pressure task		Writing speed task	
	Strong	Weak	Fast	Careful
Pressure				
Pen pressure (g)	433 ± 92	83 ± 33**	222 ± 74	200 ± 63*
Writing surface pressure (g)	506 ± 114	83 ± 40**	248 ± 87	223 ± 76*
Hand pressure (g)	787 ± 259	350 ± 150**	432 ± 179	451 ± 176
Forearm pressure (g)	1,881 ± 730	1,888 ± 726	1,747 ± 639	1,815 ± 761
Degree of variation				
Pen pressure (ΔPi)	5,050 ± 830	4,060 ± 330**	5,040 ± 740	4,170 ± 450**
Writing surface pressure (ΔPi)	10,760 ± 3,100	9,220 ± 700**	10,690 ± 1,420	9,460 ± 1,200**
Writing speed (mm/s)	11.49 ± 0.31	11.70 ± 0.32**	21.85 ± 1.16	7.78 ± 0.12**
Range of motion				
Palmar flexion and dorsiflexion (degrees)	9.2 ± 2.5	7.7 ± 2.2**	8.1 ± 2.3	8.1 ± 2.5
Radial and ulnar flexion (degrees)	12.6 ± 3.3	11.0 ± 2.3**	11.3 ± 2.8	11.4 ± 2.8
Index finger PIP flexion (degrees)	19.0 ± 7.0	18.4 ± 6.0	18.7 ± 6.9	19.7 ± 6.2
Area of overlap (pixels)	435 ± 601	650 ± 712*	1,177 ± 1,202	229 ± 353**

Use for **Goal 1.1** to define force. We specifically picked the stronger force so that the laptop designs for the strongest case possible so that they will always support the screen rigidly.

Abstract

[Purpose] The purpose of this study was to present control data on writing pressure, the amount of weight on the upper limb, joint angle, and the area of overlap from the sample in graphic tracing tasks, and to extract factors that affect pen-operating skills. [Participants and Methods] The participants included 35 healthy volunteers who performed 1-inch graphic tracing and writing. The measurements were recorded under two conditions as follows: strong and weak writing pressure tasks, and fast and careful writing speed tasks. [Results] In the writing pressure task, increase in pen pressure was accompanied by increase in hand pressure; however, the forearm pressure did not change. In the writing speed task, no significant difference in pressure was observed, except for a slight difference in writing pressure. The degree of variation in pen pressure and areas of overlap were greater in the fast task than in the careful task. Two multiple regression models revealed the contributing factors to pen-operating skills, both showing that the degree of variation in pen pressure was significantly involved. [Conclusion] The multiple regression analysis results showed that pen-operating skills can likely be evaluated, where the degree of variation in writing pressure is an index for deciding the effect of treatment intervention.

This is an extract from source [1] which, like the figure above, puts into light the obvious variables when writing and how this influences all our objectives. Factors such as ergonomics, reliability and forces that cause such natural screen vibrations all are influenced by the speed you write and the task.



Not every kind of plastic is suitable for these tough jobs. But the ones that are can help streamline workflow, save money and complete projects more effectively than ever before.

Acrylonitrile Butadiene Styrene (ABS)

ABS plastic sheet is one of the most widely-used plastics on the market for its affordability and toughness. Other key features that make it ideal for heavy-duty plastic equipment include its incredible impact resistance and ease of machinability, requiring little power to be shaped and finished to the desired dimensions for an application. It also responds well to thermoforming, so it can easily be heated and bent into shape.

Common Applications of ABS Plastic Equipment

- Machine housings for factory equipment
- Vehicle instrument panels
- Storage cases on construction sites
- Prototyping and modeling
- Guards and covers for machines/equipment



Material properties of ABS

Polycarbonate (PC)

Polycarbonate is the toughest plastic out there, at more than 20 times stronger than acrylic and 200 times stronger than glass. It has great clarity, so it's frequently used in the types of plastic equipment that require see-through applications. Polycarbonate also has a great balance of engineering properties that give it the ability to operate in a broad range of temperatures, resist impact and maintain dimensional stability. It's also a thermoplastic, so it can be reheated and remolded into new shapes as needed.

Common Applications of HDPE Plastic Equipment

- Industrial and commercial glazing (windows)
- Transparent machine guards
- Heavy machinery windows and windshields
- Face shields and sight glasses
- Semiconductor components

Material properties of PC

increased strength.

GRADES OF FERRITIC STAINLESS STEEL AND USES INCLUDE:

- **409 stainless steel:** Automotive exhaust systems, electrical kiosks and distribution boxes
- **430 stainless steel:** Automotive trim, chimney liners, dish washer linings, fasteners, refrigerator cabinet panels and stove trim rings.
- **430LI stainless steel:** Used for appliances, architectural screens, cooking utensils, fasteners, food processing and interior trim.
- **434 stainless steel:** Architectural applications, automotive trim, dishwashers, downspouts and gutters, gas burners, heating units, oil refinery equipment, restaurant equipment, roofing and siding.
- **439 stainless steel:** Automotive exhaust manifolds, hot water tank components and residential furnaces, as well as in the chemical processing, petroleum refining and power generation industries.
- **442 stainless steel:** Food processing equipment, heavy gauge welded components, pharmaceutical processing equipment and surgical implants.
- **444 stainless steel:** Cladding and roofing in marine environments, food processing equipment, hot water heaters and stainless steel tubing for heat exchangers.
- **446 stainless steel:** Boiler components, burners, flame rods, injector nozzles for molten compounds and thermocouple protectors.

MAIN PROPERTIES OF FERRITIC STAINLESS STEEL

While heat treatment won't harden or strengthen ferritic steels, they can be cold-worked and softened through an annealing process, which involves heating the metal and then slowly cooling it. Though they're neither as strong nor as resistant to corrosion as austenitic steels, ferritic grades normally have more beneficial mechanical properties. They tend to be easy to weld, though some ferritic grades have limitations, restricting use of these steels to thinner gauges.

Because they don't contain nickel and have lower amounts of chromium, standard ferritic stainless steels normally cost less than austenitic grades. The most widely used ferritic grade of stainless steel is SS430. With lower chromium content, SS409 is less corrosion-resistant. In contrast, SS439 resists corrosion better and is more stable in high temperatures. Depending on the alloy, ferritic stainless steels exhibit several beneficial properties.

CORROSION RESISTANCE

This source extract was discussed in Appendix B but is primarily about material properties of stainless steel, the final material we used for some components subjected to severe load in the syringe design.

CIV102 Course Notes Extract:

Chapter 14 – Euler Buckling of Struts

Overview

Slender members in compression can fail suddenly due to buckling. This chapter presents the derivation of Euler's equation for predicting the load at which buckling takes place.

Members in Compression

Consider the prismatic member shown in Fig. 14.2. If the two ends of the member are subjected to tensile forces, T , the only way for the ends of the member to move apart is if the member elongates. The strains experienced by the member as it stretches can be calculated using the equation $\epsilon = \Delta l / L_0$, and the member will fail when the stress in the member, $\sigma = T / A$, is equal to the ultimate tensile strength of the material.

Consider now the members subjected to compression forces in Fig. 14.3. Under the compressive forces, the two ends of the member are forced to come together. However, unlike the member in tension, there are two possible ways for the member to deform to allow this to happen. The first way is for the member to simply shorten, which is the opposite of what would happen if it was instead in tension. The second way is if the member, instead of changing length, curves to bring the two ends together. These two actions are shown on the central and right figures in Fig. 14.3 respectively.

Failure due to the first mode of deformation, which typically occurs for short, stocky members, is called **crushing**, and the force which causes crushing is sometimes referred to as the **squash load**. The second mode of failure, which commonly occurs in long, slender members, is called **buckling**. How a member fails depends on the relative amount of force required to cause crushing or buckling; whichever is easier will be the determining cause of failure.

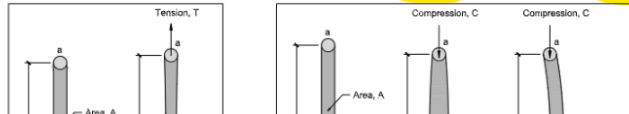


Fig. 14.1 – Soviet-era stamp celebrating the 250th birthday of the famous mathematician Leonard Euler.

Calculating the squash load of a member, P_{crush} , is straightforward if the cross-sectional area A and the ultimate compressive stress, σ_{crush} , are known:

$$P_{crush} = \sigma_{crush} A \quad (14.1)$$

Calculating the load which causes buckling to take place is more challenging because failure involves the member bending. The solution to the buckling problem was eventually solved by Leonhard Euler in 1757, leading to his celebrated equation for the Euler load, or the load causing buckling, P_e :

$$P_e = \frac{\pi^2 EI}{L^2} \quad (14.2)$$

Derivation of the Euler Load for Elastic Buckling

Euler's derivation of the buckling equation begins with the following assumptions on the member, shown in Fig. 14.4, which has a length of L and is being subjected to a compression force P which causes the member to curve as it buckles:

- The material is homogenous and linear elastic, having a uniform Young's modulus, E , and second moment of area, I .
- The top and bottom ends of the member are free to rotate. Furthermore, the top of the member is free to move vertically, and the bottom of the member is translationally fixed in place.
- The member is initially perfectly straight and either end is free to translate horizontally.

Note: Recall that the product EI is the flexural stiffness of a member.



These few extracts explain the concept of buckling, a key aspect of our splartz that we used FEA to analyze. Particularly in the second figure, we used the equation that was derived above to have an intuitive understanding of what we need to change to prevent failure. We noticed that the E value - material dependent is the only aspect we could change as CAD design was finalized. This resulted in us iterating through materials to prevent buckling.

Table 2. Average APF masures for all conditions, in [g] unless otherwise stated. Note the maximum variance and positive time trend in the cursive script condition (H). APF(0) and APF(n) denote linearly estimated initial and final force level, be is the gain of the time trend, r is correlation of APF vs time

	mean	var[g ²]	APF(0)	APF(n)	b[g/s]	r[]
S	83.6	284.5	91.3	75.9	-2	-0.27
W	87.5	250.6	95.0	80.1	-1	-0.29
C+	108.9	295.9	116.0	101.8	-1	-0.25
H	109.7	806.4	91.6	127.7	+8	+0.39
F	113.5	259.2	121.2	105.8	-1	-0.26
C-	115.3	307.7	124.0	106.6	-2	-0.29

Table 3. Average first-order (Pearson) correlations over subjects (N = 16) between APF and finger displacements S_f and between APF and wrist displacement S_w for all conditions

Effector:	Condition:	F	W	C+	C-	S	H	Mean
$R(APF, S_f)$ (Fingers)		-0.36	-0.09	-0.39	-0.24	-0.20	-0.22	-0.23
$R(APF, S_w)$ (Wrist)		-0.02	-0.15	-0.33	-0.11	-0.14	-0.10	-0.14
Mean		-0.20	-0.03	-0.36	-0.0	-0.17	-0.17	-0.16

Average pressures exerted by different gestures on a screen.

